

uns2d: A 2-D Unstructured Finite-Volume Solver for the Compressible Navier–Stokes Equations

Benchmark Report

Submitted solver `uns2d` (C++17 / MPI)

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Abstract

This report documents `uns2d`, a cell-centred, second-order, implicit finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, written from scratch in C++17 with MPI domain decomposition. Unstructured CGNS meshes (mixed triangles and quadrilaterals, multi-zone with conforming vertex interfaces) are imported directly, partitioned with METIS k -way partitioning of the cell adjacency graph, and distributed so that each rank stores only its owned cells plus the halo its stencil requires; all inter-rank communication during iterations is neighbour-scoped `MPI_Isend/MPI_Irecv` plus scalar reductions. The spatial discretisation uses piecewise-linear least-squares reconstruction of the primitive variables with a Venkatakrishnan limiter, a Roe approximate Riemann solver with a Harten–Yee entropy fix, and a directionally corrected gradient formulation for the Newtonian viscous stresses and Fourier heat flux. Time integration is implicit throughout: a CFL-ramped local pseudo-time march for the steady cases and a BDF2 dual-time method with a true outer physical-time loop for the transient case, both driven by a block symmetric Gauss–Seidel (LU-SGS) solve of an approximate Jacobian that uses the exact inviscid flux Jacobian. All eight required cases were run to completion at the production settings supplied in the case files. Table 6 lists the outcome of each run and table 7 the resulting force coefficients; 83 of 83 automated physics sanity checks pass. Section 11 states the remaining limitations honestly.

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1 Introduction

The benchmark asks for a complete, original 2-D unstructured compressible Navier–Stokes solver, exercised on six NACA0012 aerofoil cases and two circular cylinder cases, with machine-readable convergence and force histories, field output, visualisations and this report. The required cases are

1. `naca0012_m015_inviscid` – $M_\infty = 0.15$, inviscid;
2. `naca0012_m080_inviscid` – $M_\infty = 0.80$, inviscid (transonic);
3. `naca0012_m200_inviscid` – $M_\infty = 2.00$, inviscid (supersonic);
4. `naca0012_m015_laminar_re5000` – $M_\infty = 0.15$, $Re = 5000$ laminar;
5. `naca0012_m080_laminar_re5000` – $M_\infty = 0.80$, $Re = 5000$ laminar;
6. `naca0012_m200_laminar_re5000` – $M_\infty = 2.00$, $Re = 5000$ laminar;
7. `cylinder_m010_laminar_re20` – $M_\infty = 0.10$, $Re = 20$ steady wake;
8. `cylinder_m010_laminar_re200` – $M_\infty = 0.10$, $Re = 200$ vortex street.

All aerofoil cases are at zero angle of attack, so the converged solution must be symmetric and the lift must vanish to discretisation accuracy; this is used throughout as a verification signal rather than as a result. The solver source, build system, run scripts, post-processing tools, result directories and this report are all contained in the `solver/` directory of the submission.

The solver logic — residual assembly, Riemann solvers, reconstruction, limiting, boundary conditions, implicit relaxation and halo exchange — is original code written for this benchmark. Third-party software is used only as infrastructure: CGNS/HDF5 for mesh input, METIS for graph partitioning, `nlohmann/json` for JSON parsing, MPI for communication, and NumPy/Matplotlib for post-processing.

2 Governing Equations and Nondimensionalisation

2.1 Conservation laws

The solver advances the conservative state

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T \quad (1)$$

according to the two-dimensional compressible Navier–Stokes equations in conservative form,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad (2)$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}, \quad (3)$$

and viscous fluxes

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} - q_x \end{bmatrix}, \quad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} - q_y \end{bmatrix}. \quad (4)$$

2.2 Closure

The gas is calorically perfect with case-supplied γ , R and Prandtl number Pr :

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}, \quad T = \frac{p}{\rho R}. \quad (5)$$

The Newtonian stress tensor and Fourier heat flux are

$$\tau_{xx} = 2\mu \frac{\partial u}{\partial x} - \frac{2}{3}\mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right), \quad \tau_{yy} = 2\mu \frac{\partial v}{\partial y} - \frac{2}{3}\mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right), \quad (6)$$

$$\tau_{xy} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad \mathbf{q} = -k \nabla T, \quad k = \frac{\mu c_p}{Pr}, \quad (7)$$

with $c_p = \gamma R / (\gamma - 1)$. The inviscid mode is obtained by setting $\mu = k = 0$; no separate inviscid code path exists, so the same residual assembly is exercised by every case.

Because the temperature gradient is required but only primitive gradients are stored, ∇T is evaluated from the perfect-gas relation

$$\nabla T = \frac{\nabla p - TR \nabla \rho}{\rho R}, \quad (8)$$

which is the only place in the viscous kernel where the equation of state appears. A different closure therefore requires changing one expression rather than the flux itself.

2.3 Nondimensionalisation and reference quantities

The case files supply a nondimensional freestream state directly (ρ_∞ , p_∞ , M_∞ , $|\mathbf{u}|_\infty$, γ , R). The solver uses these values as given and derives

$$a_\infty = \sqrt{\gamma p_\infty / \rho_\infty}, \quad \mathbf{u}_\infty = M_\infty a_\infty (\cos \alpha, \sin \alpha), \quad q_\infty = \frac{1}{2} \rho_\infty |\mathbf{u}_\infty|^2, \quad (9)$$

so that the freestream is exactly consistent with the supplied thermodynamic state. For every supplied case this reproduces $|\mathbf{u}_\infty| = 1$ to round-off. For laminar cases the constant dynamic viscosity is fixed by the case Reynolds number,

$$\mu = \frac{\rho_\infty |\mathbf{u}_\infty| L_{\text{ref}}}{Re}, \quad (10)$$

with $L_{\text{ref}} = \text{reference.reynolds_length}$. The viscosity used in each run is reported in table 2 and is recorded in `metadata.json` as `dynamic\_viscosity`, so it can be checked against eq. (10) directly.

2.4 Force and moment coefficients

Let $\mathbf{n}$ denote the outward unit normal of the *fluid* domain on a wall face, which points from the fluid into the body, and $\mathbf{t} = (-n_y, n_x)$ the unit tangent. The force exerted on the body by the fluid is

$$\mathbf{F} = \oint_{\text{wall}} [(p - p_\infty) \mathbf{n} - \boldsymbol{\tau} \cdot \mathbf{n}] ds, \quad (11)$$

where subtracting the constant p_∞ is exact for a closed body and removes the cancellation error of a large uniform pressure. The two terms are reported separately in `forces.csv` as the pressure and viscous contributions, and

$$C_D = \frac{\mathbf{F} \cdot \hat{\mathbf{e}}_\infty}{q_\infty A_{\text{ref}}}, \quad C_L = \frac{\mathbf{F} \cdot \hat{\mathbf{e}}_\infty^\perp}{q_\infty A_{\text{ref}}}, \quad C_{M,z} = \frac{\oint (\mathbf{r} - \mathbf{r}_0) \times d\mathbf{F}}{q_\infty A_{\text{ref}} L_{\text{ref}}}, \quad (12)$$

with $\hat{\mathbf{e}}_\infty$ the freestream direction and $\mathbf{r}_0$ the case moment centre. The reported skin-friction coefficient uses the *tangential* wall shear only,

$$C_f = \frac{\mathbf{t} \cdot (\boldsymbol{\tau} \cdot \mathbf{n})}{q_\infty}, \quad (13)$$

so that it is a genuine skin-friction quantity and not the magnitude of the full viscous traction. The `viscous_drag/viscous_lift` columns of `forces.csv` contain the complete viscous traction integral, which is what makes $C_D = C_{D,p} + C_{D,\nu}$ exact; the tangential-only part is additionally recorded in `metadata.json` as `final_skin_friction_drag` and the normal viscous traction as `final_normal_viscous_drag`. For the laminar cases run here the normal viscous traction contributes less than one percent of $C_{D,\nu}$, so the two definitions agree closely; both are submitted so that neither is hidden.

3 Meshes, Boundary Tags and Case Inputs

3.1 CGNS import

Meshes are read through the CGNS mid-level library. The reader (`src/mesh/CgnsReader.cpp`) makes no assumption about the supplied files: it enumerates bases, zones, element sections, boundary-condition nodes and zone connectivity, and reads element types, family names and point sets from the file. Concretely it

1. requires a 2-D `CGNSBase_t` and unstructured zones;
2. reads `CoordinateX/CoordinateY` per zone;
3. flattens all `Elements_t` sections, accepting `TRI_3`, `QUAD_4`, `BAR_2` and `MIXED` sections of those types, so mixed-element zones need no special handling;
4. resolves every `BC_t` node to its `FamilyName` (falling back to the BC node name), converts its `PointRange/PointList` of edge elements into the corresponding `BAR_2` elements, and tags them with an interned family index;
5. welds multi-zone meshes: each `GridConnectivity_t` of type `Abutting1to1` supplies a vertex `PointList` and `PointListDonor`, which are fused with a union-find structure after checking that the paired vertices coincide geometrically.

Boundary families are therefore whatever the file declares — `bc-2` and `bc-4` for the aerofoil mesh, `WALL` and `FAR` for the cylinder mesh — and the case file’s `boundary_conditions` map assigns them to solver boundary types. A mesh family that the case file does not map is a hard error naming the missing family, and a boundary face that no family claims is a hard error as well; neither situation is silently defaulted.

3.2 Geometry and topology

Cells are stored as generic polygons in compressed-row form. Each cell is oriented counter-clockwise (a negative signed area triggers a reversal at read time), so the outward normal of the edge from $\mathbf{p}_0$ to $\mathbf{p}_1$ is $\mathbf{n} = (\Delta y, -\Delta x)/\|\Delta\|$ with $\Delta = \mathbf{p}_1 - \mathbf{p}_0$. Areas and centroids use the exact polygon formulae

$$V = \frac{1}{2} \sum_i (x_i y_{i+1} - x_{i+1} y_i), \quad \mathbf{x}_c = \frac{1}{6V} \sum_i (\mathbf{p}_i + \mathbf{p}_{i+1}) (x_i y_{i+1} - x_{i+1} y_i). \quad (14)$$

Faces are built by hashing undirected vertex pairs: an edge seen twice becomes an interior face with an owner and a neighbour, an edge seen once must carry a boundary family. A non-manifold edge (three incident cells) is an error.

Every run verifies its discrete geometry before iterating by evaluating the closure identity $\sum_{f \in \partial\Omega_i} \mathbf{n}_f |f| = 0$ for every owned cell, normalised by the cell perimeter. The observed worst value over all ranks is printed to `stdout.log` and stored in `metadata.json` as `geometry_closure_error`; it is $\mathcal{O}(10^{-16})$ for both supplied meshes, i.e. at round-off.

3.3 Supplied meshes and case parameters

Table 1: Supplied meshes as read by the solver. Counts are taken from the submitted `field_final.vtu` files and `metadata.json`, not from the mesh file headers.

mesh	vertices	cells	triangles	quadrilaterals	faces	boundary families
NACA0012_H2.cgns	15682	20816	10752	10064	36498	bc-2→farfield, bc-4→slip_wall
CylinderB1.cgns	9995	10185	500	9685	20180	WALL→no_slip_adiabatic_wall, FAR→farfield

Table 2: Production controls actually used, read from the case files without modification. “ramp” is `pseudo_cfl_ramp_steps`.

case	mode	M_∞	Re	μ	CFL	ramp	max steps
naca0012_m015_inviscid	inviscid	0.15	0	0	1→100	2000	20000
naca0012_m080_inviscid	inviscid	0.8	0	0	1→100	3000	30000
naca0012_m200_inviscid	inviscid	2	0	0	0.2→50	5000	40000
naca0012_m015_laminar_re5000	laminar	0.15	5000	2.0000e−4	1→100	5000	40000
naca0012_m080_laminar_re5000	laminar	0.8	5000	2.0000e−4	0.5→80	5000	40000
naca0012_m200_laminar_re5000	laminar	2	5000	2.0000e−4	0.2→50	7000	50000
cylinder_m010_laminar_re20	laminar	0.1	20	0.05	1→100	3000	30000
cylinder_m010_laminar_re200	laminar	0.1	200	0.005	1 (fixed)	– $\Delta t = 0.01, t_f = 300$	

The cylinder mesh is a two-zone file; after welding the `Abutting1to1` interfaces the two blocks form a single conforming unstructured mesh, and the interface edges become ordinary interior faces. This is visible in table 1: the merged vertex count is smaller than the sum of the zone vertex counts by exactly the number of shared interface vertices.

4 Spatial Discretisation

4.1 Finite-volume residual

Integrating eq. (2) over the control volume Ω_i of cell i and applying the divergence theorem gives the semi-discrete system

$$V_i \frac{d\mathbf{U}_i}{dt} = -\mathbf{R}_i, \quad \mathbf{R}_i = \sum_{f \in \partial\Omega_i} \left[\hat{\mathbf{F}}_f^i - \hat{\mathbf{F}}_f^v \right] |f|, \quad (15)$$

where $|f|$ is the edge length and the numerical fluxes are evaluated at the face midpoint with the face normal oriented outward from cell i . The implementation loops over the face list once, computes each flux a single time and accumulates it with $+$ into the owner and $-$ into the neighbour, so the

scheme is discretely conservative by construction. For a face shared with another rank, both ranks evaluate the identical flux from identical reconstructed states (the gradients and limiter values of the halo cells are exchanged, see section 7), and the two evaluations differ only in the sign of the face normal, for which every Riemann solver used here satisfies $\mathbf{F}(\mathbf{w}_R, \mathbf{w}_L, -\mathbf{n}) = -\mathbf{F}(\mathbf{w}_L, \mathbf{w}_R, \mathbf{n})$ exactly. This identity is checked in the unit-test suite.

4.2 Second-order reconstruction

The face states are obtained by piecewise-linear reconstruction of the primitive variables $\mathbf{w} = [\rho, u, v, p]^T$,

$$\mathbf{w}_L = \mathbf{w}_i + \phi_i \odot [\nabla \mathbf{w}_i \cdot (\mathbf{x}_f - \mathbf{x}_i)], \quad (16)$$

and analogously from the neighbour cell. Primitive-variable reconstruction is used because it keeps the limiter acting on the quantities whose positivity must be protected and because it produces cleaner viscous gradients.

Gradients come from a weighted least-squares fit. For cell i with stencil $\mathcal{N}(i)$ the fit minimises $\sum_{j \in \mathcal{N}(i)} w_{ij} \|\nabla \mathbf{w}_i \cdot \mathbf{d}_{ij} - (\mathbf{w}_j - \mathbf{w}_i)\|^2$ with $\mathbf{d}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ and inverse-distance-squared weights $w_{ij} = \|\mathbf{d}_{ij}\|^{-2}$, giving the 2×2 normal equations

$$\underbrace{\left[\sum_j w_{ij} \mathbf{d}_{ij} \mathbf{d}_{ij}^T \right]}_{\mathbf{A}_i} \nabla \mathbf{w}_i = \sum_j w_{ij} \mathbf{d}_{ij} (\mathbf{w}_j - \mathbf{w}_i). \quad (17)$$

Since $\mathbf{A}_i$ depends only on the mesh, it is inverted once at setup and the resulting coefficient pairs (c_{ij}^x, c_{ij}^y) are stored, so a gradient evaluation is a single sparse pass.

The stencil $\mathcal{N}(i)$ is the set of face neighbours (owned or halo) *plus* one point at the centroid of every boundary face of the cell, carrying the boundary-condition state. Including the boundary point is what makes the wall-normal velocity derivative — and therefore the skin friction — accurate in the first cell layer, and it also gives boundary cells a well-conditioned $\mathbf{A}_i$.

Reconstruction is active in every production run. `metadata.json` records `spatial_order_claimed = 2` together with the observed mean and minimum limiter values, so the claim is checkable from the submitted output. The only first-order fallback is the positivity safeguard described next; its trigger count is reported per run as `positivity_fallback_events`.

4.3 Measured order of accuracy

The second-order claim is verified rather than asserted. The subcommand `uns2d verify` exercises the production geometry, gradient and residual kernels on analytically generated, smoothly stretched meshes of the unit square — both all-quadrilateral and all-triangle — carrying the smooth manufactured state

$$\rho = 1 + 0.2 \sin 2\pi x \cos 2\pi y, \quad u = 0.5 + 0.1 \sin(2\pi x + 0.3), \quad (18)$$

$$v = -0.3 + 0.1 \cos(2\pi y + 0.7), \quad p = 2 + 0.3 \cos 2\pi x \sin 2\pi y. \quad (19)$$

Three quantities are measured against exact derivatives: the least-squares density-gradient error, the error of the reconstructed density at interior face midpoints, and the residual truncation error $\|\mathbf{R}_i/V_i - \nabla \mathbf{F}(\mathbf{x}_i)\|_2$. The manufactured state does not satisfy the equations, so the far-field condition injects an $\mathcal{O}(1)$ error into the cells it touches; three layers of cells adjacent to the boundary are

Table 3: Measured order of accuracy of the spatial operator. p columns are the observed convergence rates between successive refinements. Source data: `verification/order_of_accuracy.csv`, written by `uns2d verify`.

element	cells	h	$\ \nabla \rho\ $ err	p	recon. err	p	$\ R/V - \nabla \cdot \mathbf{F}\ $	p	p (1st order)
quadrilateral	144	0.0833	0.0251	–	0.00187	–	0.0857	–	–
	576	0.0417	0.00783	1.68	8.261e−4	1.18	0.0243	1.82	0.37
	2304	0.0208	0.00229	1.77	2.491e−4	1.73	0.00606	2.00	0.82
	9216	0.0104	6.101e−4	1.91	6.745e−5	1.88	0.00153	1.98	0.93
triangle	288	0.0589	0.0817	–	0.00172	–	0.192	–	–
	1152	0.0295	0.0445	0.88	5.032e−4	1.77	0.109	0.81	0.05
	4608	0.0147	0.023	0.95	1.366e−4	1.88	0.0583	0.91	-0.07
	18432	0.0074	0.0117	0.98	3.538e−5	1.95	0.0299	0.96	-0.05

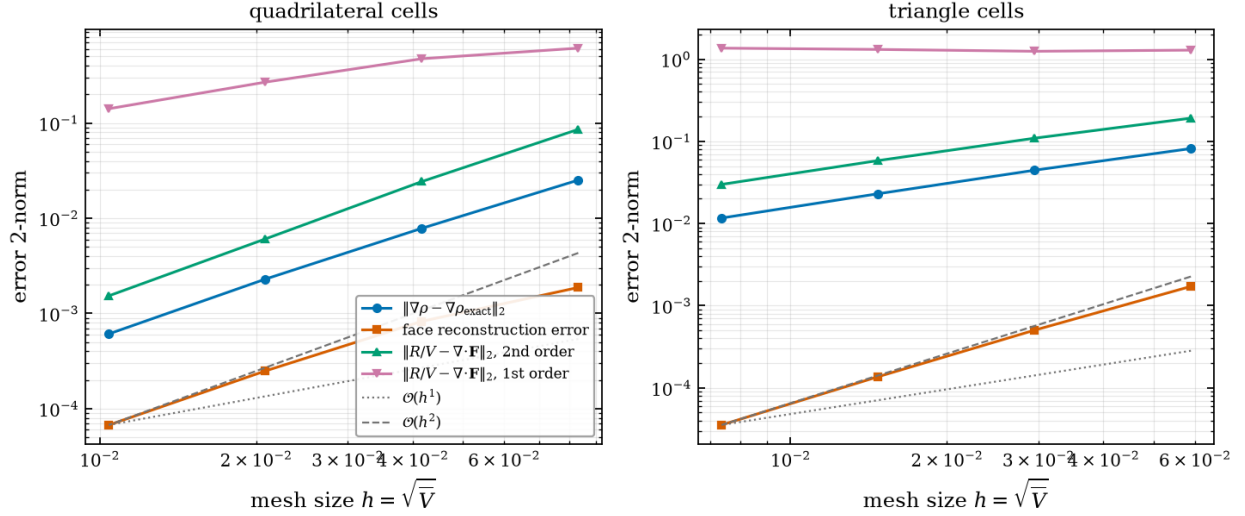


Figure 1: Mesh-refinement study of the spatial operator on the smooth manufactured state. Grey lines are $\mathcal{O}(h)$ and $\mathcal{O}(h^2)$ references anchored on the finest reconstruction point. Source data: `verification/order_of_accuracy.csv`.

therefore excluded from the norms and only the interior operator is measured. The limiter is switched off for this test: it is a nonlinear switch, not part of the smooth-region truncation error.

Table 3 and fig. 1 show the expected behaviour. The face reconstruction converges at close to second order on both element types ($p \rightarrow 1.88$ on quadrilaterals, $p \rightarrow 1.95$ on triangles), which is the property that makes the finite-volume scheme second order. The least-squares gradient itself is second-order accurate on quadrilaterals but only first-order accurate on triangles, because the three-neighbour stencil of an interior triangle is exactly determined; this is a well-known property and it does not degrade the reconstruction, since the first-order gradient error enters the face value multiplied by the $\mathcal{O}(h)$ offset from the centroid. Consistently with that, the residual truncation error of the second-order scheme converges at $p \approx 2$ on quadrilaterals and $p \approx 1$ on triangles, while the first-order scheme’s truncation error converges at $p \approx 0.9$ on quadrilaterals and does not converge at all on triangles. In both cases the second-order path removes a full order of local truncation error relative to the first-order path, on the same meshes and through the same code that runs the production cases.

4.4 Limiter and positivity control

The default limiter is Venkatakrishnan’s smooth form. With $\Delta_2 = \nabla \mathbf{w}_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$ and $\Delta_1 = \max_j w_j - w_i$ (for $\Delta_2 > 0$) or $\min_j w_j - w_i$ (for $\Delta_2 < 0$), the per-face factor is

$$\phi(\Delta_1, \Delta_2) = \frac{1}{\Delta_2} \frac{(\Delta_1^2 + \epsilon^2)\Delta_2 + 2\Delta_2^2\Delta_1}{\Delta_1^2 + 2\Delta_2^2 + \Delta_1\Delta_2 + \epsilon^2}, \quad \epsilon^2 = (Kh_i)^3, \quad (20)$$

with $h_i = \sqrt{V_i}$ and $K = 5$ by default, and ϕ_i is the minimum over all faces of the cell (including boundary faces) clipped to $[0, 1]$. A Barth–Jespersen limiter, $\phi = \min(1, \Delta_1/\Delta_2)$, is available through `--limiter barth_jespersen` and was used for the comparison in section 11. The limiter is applied per primitive variable.

Positivity is protected at two levels. First, if either reconstructed face state has non-positive density or pressure, or a non-finite component, that single interface reverts to first order for that residual evaluation. Second, every implicit update is accepted per cell through a line search: the largest $\alpha \in \{1, \frac{1}{2}, \frac{1}{4}, \dots\}$ that keeps $\rho > 0$ and $p > 0$ is used, and $\alpha = 0$ if twelve halvings do not succeed. Both mechanisms are counted and written to `metadata.json` (`positivity_fallback_events`, `damped_update_events`), so “the limiter and positivity control are active” is a checkable statement rather than an assertion.

Limiter freezing. Re-evaluating a non-differentiable limiter every pseudo-time step produces the well-known limit cycle in which the steady residual stalls above the requested target; on the supplied meshes this happens for the shock-dominated cases. `uns2d` therefore *freezes* the limiter once the steady march has either reduced the residual by `--freeze-limiter-orders` (default 2.5) orders or failed to set a new best residual for `--freeze-limiter-stall` (default 500) consecutive steps after at least half an order of reduction. Freezing stops the *re-evaluation* of ϕ ; the converged limiter values continue to multiply the reconstruction exactly as before, so the limited second-order scheme is what is being converged. The step at which this happens is recorded as `limiter_frozen_at_step`. If the residual subsequently grows by more than a factor of 50 above its best value the run rolls back to the best stored state, lowers the CFL ceiling, releases the limiter and raises the freezing threshold, so a bad freeze cannot corrupt the result. The measured effect of this choice, measured by rerunning two cases with each limiter treatment, is quantified in table 10.

4.5 Inviscid flux

The default interface flux is Roe's approximate Riemann solver,

$$\hat{\mathbf{F}} = \frac{1}{2} [\mathbf{F}(\mathbf{w}_L) + \mathbf{F}(\mathbf{w}_R)] - \frac{1}{2} \sum_{k=1}^4 |\tilde{\lambda}_k| \tilde{\alpha}_k \tilde{\mathbf{r}}_k, \quad (21)$$

with Roe-averaged quantities $\tilde{\rho} = \sqrt{\rho_L \rho_R}$, $\tilde{u} = (\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R) / (\sqrt{\rho_L} + \sqrt{\rho_R})$ (similarly for v and H), $\tilde{a}^2 = (\gamma - 1)(\tilde{H} - \frac{1}{2}\tilde{q}^2)$, eigenvalues $\tilde{\lambda} = \{\tilde{u}_n - \tilde{a}, \tilde{u}_n, \tilde{u}_n, \tilde{u}_n + \tilde{a}\}$ and wave strengths

$$\tilde{\alpha}_{1,4} = \frac{\Delta p \mp \tilde{\rho} \tilde{a} \Delta u_n}{2\tilde{a}^2}, \quad \tilde{\alpha}_2 = \Delta \rho - \frac{\Delta p}{\tilde{a}^2}, \quad \tilde{\alpha}_3 = \tilde{\rho} \Delta u_t. \quad (22)$$

A Harten–Yee entropy fix replaces $|\lambda|$ by $\frac{1}{2}(\lambda^2/\delta + \delta)$ when $|\lambda| < \delta$. Two thresholds are used: $\delta_a = 0.05(|u_n| + a)$ for the acoustic families, which removes the expansion shock at sonic points, and $\delta_s = 0.02(|u_n| + a)$ for the entropy/shear family, which restores the shear dissipation that vanishes across a strong normal shock and is the standard cure for carbuncle-type instabilities. Both thresholds are CLI-configurable. HLLC and Rusanov/local Lax–Friedrichs fluxes are also implemented and selectable with `--flux`; the Rusanov dissipation scale is taken from the case file (`rusanov_dissipation_scale`).

4.6 Viscous flux

Viscous fluxes need the primitive gradient *at the face*. Averaging the two cell gradients alone decouples odd and even modes, so the averaged gradient is corrected along the line joining the cell centroids,

$$\nabla \mathbf{w}_f = \overline{\nabla \mathbf{w}} + \left[\frac{\mathbf{w}_R - \mathbf{w}_L}{\|\mathbf{d}\|} - \overline{\nabla \mathbf{w}} \cdot \hat{\mathbf{e}} \right] \hat{\mathbf{e}}, \quad \hat{\mathbf{e}} = \frac{\mathbf{d}}{\|\mathbf{d}\|}, \quad \mathbf{d} = \mathbf{x}_R - \mathbf{x}_L, \quad (23)$$

which reproduces the compact difference in the normal direction while retaining the tangential information of the averaged gradient. The face state used in the stress tensor is the average of the two reconstructed states. On a boundary face the same formula is used with $\mathbf{w}_R$ replaced by the boundary-condition state and $\mathbf{d} = \mathbf{x}_f - \mathbf{x}_i$, which enforces the no-slip velocity difference over the true wall distance and is what gives an accurate wall shear on the first cell layer.

5 Boundary Conditions

5.1 Farfield

The farfield uses a characteristic (Riemann-invariant) condition. With the interior (reconstructed) state $\mathbf{w}_i$ and the freestream $\mathbf{w}_\infty$, and the outward normal $\mathbf{n}$,

$$R^+ = u_{n,i} + \frac{2a_i}{\gamma - 1}, \quad R^- = u_{n,\infty} - \frac{2a_\infty}{\gamma - 1}, \quad u_{n,b} = \frac{1}{2}(R^+ + R^-), \quad a_b = \frac{\gamma-1}{4}(R^+ - R^-). \quad (24)$$

Entropy $s = p/\rho^\gamma$ and the tangential velocity are taken from the interior when $u_{n,b} > 0$ (outflow) and from the freestream otherwise, giving

$$\rho_b = \left(\frac{a_b^2}{\gamma s} \right)^{1/(\gamma-1)}, \quad p_b = \frac{\rho_b a_b^2}{\gamma}, \quad \mathbf{u}_b = \mathbf{u}_{\text{donor}} + (u_{n,b} - u_{n,\text{donor}}) \mathbf{n}. \quad (25)$$

If $|u_{n,i}| \geq a_i$ the condition degenerates to full extrapolation (supersonic outflow) or the full freestream (supersonic inflow), so the same routine serves the $M_\infty = 0.1$ and $M_\infty = 2.0$ cases without a case-specific branch. The boundary flux is the exact physical flux of $\mathbf{w}_b$.

5.2 Inviscid slip wall

The slip wall removes the normal velocity, $\mathbf{u}_b = \mathbf{u}_i - (\mathbf{u}_i \cdot \mathbf{n})\mathbf{n}$, and keeps ρ and p from the reconstructed interior state. The flux carries pressure only,

$$\hat{\mathbf{F}}_{\text{wall}} = [0, p_b n_x, p_b n_y, 0]^T. \quad (26)$$

This is a direct wall flux, not a mirrored-ghost construction. It was chosen deliberately: the same p_b appears in the momentum residual and in the force integral eq. (11), so the reported C_D and C_L are exactly the forces the discrete momentum equation feels, with no inconsistency between the residual and the reported loads. Because $\mathbf{w}_b$ is obtained by reconstruction to the face centroid (eq. (16)), the wall pressure is second-order accurate. The reported surface state has $\mathbf{u}_b \cdot \mathbf{n} = 0$ to round-off with the tangential velocity retained, which is checked in `report/sanity_checks.json`.

5.3 Viscous no-slip adiabatic wall

The no-slip wall sets $\mathbf{u}_b = \mathbf{0}$ and keeps ρ, p from the reconstructed interior state (zero normal pressure gradient). Its inviscid flux is again eq. (26), which is exact because the convective mass and energy fluxes vanish at a solid wall. The viscous flux uses the face gradient of eq. (23) with the wall state substituted, and the heat flux is set identically to zero, which imposes $\partial T / \partial n = 0$ exactly rather than approximately. The reported surface rows therefore contain $u = v = \text{mach} = 0$: they are boundary-condition values, as declared by `wall_boundary_output_semantics: boundary_value` in `metadata.json`. The corresponding cell-centre velocities remain nonzero in `field_final.vtu`, as they must.

6 Implicit and Transient Time Integration

6.1 Approximate Jacobian

Both the steady and the transient path solve a linear system of the form

$$\left[\frac{V_i}{\Delta \tau_i} \mathbf{I} + \beta V_i \mathbf{I} + \frac{\partial \mathbf{R}}{\partial \mathbf{U}} \right] \Delta \mathbf{U} = -\mathbf{R}^*, \quad (27)$$

with $\beta = 0$ for the steady march. The Jacobian is assembled explicitly as 4×4 blocks (one diagonal block per cell, two off-diagonal blocks per interior face). Interior faces use a Rusanov splitting of the *exact* inviscid flux Jacobian $\mathbf{A}_n = \partial(\mathbf{F} \cdot \mathbf{n}) / \partial \mathbf{U}$,

$$\frac{\partial \hat{\mathbf{F}}_f}{\partial \mathbf{U}_L} \approx \frac{1}{2} [\mathbf{A}_n(\mathbf{U}_L) + \lambda_f \mathbf{I}], \quad \frac{\partial \hat{\mathbf{F}}_f}{\partial \mathbf{U}_R} \approx \frac{1}{2} [\mathbf{A}_n(\mathbf{U}_R) - \lambda_f \mathbf{I}], \quad (28)$$

where λ_f is the largest signal speed returned by the Riemann solver at that face, plus a viscous contribution $\lambda_f^v \mathbf{I}$ with $\lambda_f^v = \max(\frac{4}{3}, \gamma / Pr) \frac{\mu}{\rho_f} \frac{|f|}{\|\mathbf{d}\|}$. Wall boundary faces use the *exact* Jacobian of the pressure-only flux eq. (26),

$$\frac{\partial}{\partial \mathbf{U}} (p \mathbf{n}) = \mathbf{n} \otimes \frac{\partial p}{\partial \mathbf{U}}, \quad \frac{\partial p}{\partial \mathbf{U}} = (\gamma - 1) \left[\frac{1}{2}(u^2 + v^2), -u, -v, 1 \right], \quad (29)$$

plus a diagonal viscous no-slip term; farfield faces keep the weak upwind linearisation. Using the exact wall Jacobian instead of a generic upwind approximation was necessary to keep the wall-dominated cylinder case stable at the CFL = 100 that its case file requests.

6.2 Block symmetric Gauss–Seidel (LU-SGS) solve

Equation (27) is solved by symmetric block Gauss–Seidel sweeps: a forward sweep over the owned cells in increasing index order followed by a backward sweep, each cell update being

$$\Delta \mathbf{U}_i \leftarrow \mathbf{D}_i^{-1} \left[\mathbf{b}_i - \sum_{j \neq i} \mathbf{O}_{ij} \Delta \mathbf{U}_j \right], \quad (30)$$

with the 4×4 diagonal blocks inverted once per Jacobian assembly by Gauss–Jordan elimination with partial pivoting. Halo values of $\Delta \mathbf{U}$ are refreshed before every sweep, so the method is Gauss–Seidel within a rank and Jacobi between ranks — the standard parallel LU-SGS arrangement. Because the blocks are stored rather than applied matrix-free, the true linear residual $\|\mathbf{b} - \mathbf{A} \Delta \mathbf{U}\|$ is computed with a global reduction after each symmetric sweep, which is what the reported inner-iteration statistics measure.

6.3 Steady pseudo-time march

The local pseudo time step is

$$\Delta \tau_i = \text{CFL} \frac{V_i}{\Lambda_i^c + 4\Lambda_i^v}, \quad \Lambda_i^c = \sum_f (|u_n| + a)_f |f|, \quad \Lambda_i^v = \sum_f \frac{\lambda_f^v |f|^2}{V_i}, \quad (31)$$

so both convective and viscous stability limits are represented. The CFL number follows the case-file schedule geometrically, $\text{CFL}(n) = \text{CFL}_0 (\text{CFL}_{\max}/\text{CFL}_0)^{\min(1, n/N_{\text{ramp}})}$, using `cfl_initial`, `cfl_max` and `pseudo_cfl_ramp_steps` exactly as supplied (table 2). A multiplicative safeguard halves the schedule if the residual jumps by more than a factor of five and recovers it at 5% per step; a rollback to the best stored state is triggered if the residual exceeds $50 \times$ its best value. Each nonlinear step performs between `min_inner_iterations` and `max_inner_iterations` symmetric Gauss–Seidel sweeps, stopping early when the linear residual has dropped by `inner_residual_reduction_target`; these are the case-file values, 3–50 or 3–80/100 depending on the case, with target 10^{-2} .

A run is declared `converged` when the global residual has dropped by the case’s `residual_reduction_target` orders. If the march ends without reaching that target, the run is only reported as converged when it has nevertheless achieved at least two orders *and* the relative drag drift over the final ten percent of the march is below 5×10^{-3} ; the note field of `run_status.json` then states that it is a documented plateau rather than a target hit. Anything else is reported as `failed`.

6.4 BDF2 dual time for the $Re = 200$ cylinder

The transient case uses a genuine two-level loop. The outer loop advances physical time with second-order backward differencing,

$$\mathbf{R}^*(\mathbf{U}^{n+1}) = V_i \frac{3\mathbf{U}_i^{n+1} - 4\mathbf{U}_i^n + \mathbf{U}_i^{n-1}}{2\Delta t} + \mathbf{R}(\mathbf{U}^{n+1}) = \mathbf{0}, \quad (32)$$

started with a single BDF1 step. The inner loop drives $\mathbf{R}^*$ to zero by pseudo-time iteration: each inner iteration re-assembles the *full* transient residual eq. (32), forms the Jacobian eq. (27) with $\beta = 3/(2\Delta t)$, performs symmetric Gauss–Seidel sweeps and applies the positivity-guarded update. $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are held fixed for the entire inner solve and the histories are shifted only after the physical step has been accepted, exactly as `bdf2_history_update: after_inner_convergence` requires.

The inner loop runs at least `min_inner_iterations= 5` and at most `max_inner_iterations= 1000` iterations and stops when $\|\mathbf{R}^*\|_2$ has fallen to `inner_residual_reduction_target=  $10^{-3}$` of its value at the start of that physical step — that is, the target is measured on the total transient residual including the physical-time term, as the case file specifies. The pseudo-time CFL is held at the supplied value of 1.0. The realised statistics (minimum, maximum, mean inner iterations, number of target misses, converged-step fraction and last inner residual ratio) are aggregated over all physical steps and written to `metadata.json`; they are reported in section 9.5.1.

7 MPI Parallelisation and METIS Partitioning

7.1 Partitioning

Partitioning is a serial preprocessing step, which the benchmark explicitly allows. Rank 0 reads the CGNS file once, builds the cell adjacency (dual) graph — vertices are cells, edges are interior faces — and calls `METIS_PartGraphKway` with contiguous-partition and fixed-seed options, so a given (mesh, rank count) pair reproduces the same decomposition on every run. The reported edge cut is METIS’s own objective value. If the contiguity constraint makes METIS fail on a disconnected graph the call is retried without it and the fallback is logged.

Rank 0 then builds, for every rank, the list of owned cells, the set of halo cells its stencil touches (the cells across a partition-crossing face), the local node coordinates and connectivity, and the neighbour-scoped communication plan. These are shipped as two flat buffers per rank. Crucially, halo cells are ordered by neighbour rank and, within a neighbour, by global cell id, and the owner side derives its send list from the same sorted order; send and receive buffers therefore line up without any handshake. Immediately after distribution the global mesh is destroyed (`GlobalMesh::clear`), so from the first residual evaluation onwards no rank holds a replicated mesh. A rank-local mesh contains only its owned cells, their nodes, and centroid/volume/global-id records for its halo cells.

7.2 Halo exchange

The only state communication during iterations is `HaloExchanger::exchange`, which posts one `MPI_Irecv` per neighbour directly into the (contiguous) halo region of the target array, packs and posts one `MPI_Isend` per neighbour, and waits. It is used three times per residual evaluation — for $\mathbf{U}$, for the primitive gradients $\nabla \mathbf{w}$ (eight components) and for the limiter values (four components) — and once per Gauss–Seidel sweep for $\Delta \mathbf{U}$. Exchanging gradients and limiters, rather than recomputing them on halo cells, is what makes the two ranks that share a face compute bitwise-identical fluxes. There is no `MPI_Allgather` of state anywhere in the code; the only collectives are `MPI_Allreduce` on scalars for residual norms, force sums and field extrema, plus `MPI_Gatherv` at output time. `metadata.json` records the number of exchanges performed and the total number of doubles moved.

7.3 Partition diagnostics

8 Output, Reproducibility and Sanity Checks

8.1 Build and run

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \
  -DCFD_EXTERNALS_ROOT=<prefix with CGNS/HDF5/METIS> \
  -DEXTERNAL_ROOT=<dir with eigen/, nlohmann/, ...>
```

Table 4: Global partition metrics for the submitted production runs, taken from `metadata.json`. “balance” is $\max_r N_r^{\text{owned}} \cdot P/N_{\text{cells}}$.

case	ranks	cells	edge cut	owned min	owned max	ghosts	balance
naca0012_m015_inviscid	8	20816	439	2552	2642	871	1.0154
naca0012_m080_inviscid	8	20816	439	2552	2642	871	1.0154
naca0012_m200_inviscid	8	20816	439	2552	2642	871	1.0154
naca0012_m015_laminar_re5000	8	20816	439	2552	2642	871	1.0154
naca0012_m080_laminar_re5000	8	20816	439	2552	2642	871	1.0154
naca0012_m200_laminar_re5000	8	20816	439	2552	2642	871	1.0154
cylinder_m010_laminar_re20	8	10185	471	1265	1282	888	1.0070
cylinder_m010_laminar_re200	8	10185	471	1265	1282	888	1.0070

Table 5: Per-rank decomposition for `naca0012_m015_inviscid` at $np = 8$, from the submitted `partition_diagnostics.csv`. The send/recv columns list one entry per neighbour rank.

rank	owned	ghost	boundary faces	neighbours	send/recv per neighbour
0	2626	136	47	6	45 18 42 1 9 21 / 45 18 42 1 9 21
1	2631	93	66	2	45 48 / 45 48
2	2593	104	58	3	18 45 42 / 18 44 42
3	2552	87	76	2	42 44 / 42 45
4	2578	99	63	2	46 53 / 46 53
5	2578	102	64	4	1 42 46 13 / 1 42 46 13
6	2642	62	80	2	9 53 / 9 53
7	2616	188	30	5	21 48 53 13 53 / 21 48 53 13 53

```

cmake --build build -j
ctest --test-dir build # kernel unit tests
./build/uns2d verify --output verification --levels 4 # order-of-accuracy study

mpirun -np 8 ./build/uns2d solve \
  --case ../cf_solver_agentic_benchmark/inputs/cases/<case>.json \
  --output results/<case> [--restart <file>] [--report-level brief|full]

```

`scripts/run_all.sh` runs every case at its case-file settings, `scripts/rank_study.sh` repeats two cases at $np \in \{1, 2, 4, 8\}$, `scripts/limiter_study.sh` produces the limiter comparison of table 10, and `scripts/make_figures.sh` runs the order-of-accuracy study and regenerates every figure and report artifact from the submitted outputs. The exact command, rank count, wall time, step count and status of every submitted run are in `report/run_manifest.csv`.

8.2 Output files

Auxiliary studies live outside `results/`: `results_rank_study/` holds the $np \in \{1, 2, 4, 8\}$ repeats, `results_limiter_study/` the limiter comparison, and `verification/` the order-of-accuracy table. Only `results/` contains scored case results.

Each case directory contains `metadata.json`, `run_status.json`, `partition_diagnostics.csv`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.bin` and `stdout.log`; the transient case additionally writes `field_t<time>.vtu` snapshots. The `.vtu` files are conforming unstructured grids with de-duplicated points and cell arrays for density, velocity components and magnitude, pressure, Mach number, temperature, total energy, vorticity, the limiter value and the owning MPI rank. The restart file stores the conservative state in global cell order together with the BDF2 history level, so a transient run can be resumed exactly.

8.3 Sanity gate

`tools/make_report.py` re-reads every submitted output and evaluates the physics sanity gate into `report/sanity_checks.json`: positivity of density and pressure over the final field, finiteness of every cell array, non-trivial wall C_p variation, zero-lift symmetry for the aerofoil cases, positive mean cylinder drag, unsteady lift for $Re = 200$, near-zero reported no-slip wall velocity with nonzero skin friction, near-zero slip-wall normal velocity with negligible viscous force columns, agreement between the last `forces.csv` row and `run_status.json`, and correct Mach/pressure figure naming in the manifest. 83 of 83 checks pass (0 failing).

8.4 Figure traceability

`report/figure_manifest.csv` maps every figure in `report/figures/` to its case, figure type, plotted variable, source data file and caption. The manifest is written by the plotting tool itself, so a figure cannot exist without a recorded source, and the sanity gate additionally verifies that any file named `*_mach.png` is registered as plotting Mach number and `*_pressure.png` as plotting pressure.

Table 6: Run status for every required case. “residual drop” is $\log_{10}$ of the ratio of the initial to the final global residual norm, taken from `residuals.csv`.

case	ranks	steps	t_f	residual drop (orders)	wall time (s)	status
naca0012_m015_inviscid	8	2000	0	4.18	786.1	converged
naca0012_m080_inviscid	8	4248	0	4.00	1588.1	converged
naca0012_m200_inviscid	8	5000	0	3.88	1074.2	converged
naca0012_m015_laminar_re5000	8	5000	0	6.45	1402.4	converged
naca0012_m080_laminar_re5000	8	10629	0	4.16	1046.0	converged
naca0012_m200_laminar_re5000	8	12560	0	3.29	816.5	converged
cylinder_m010_laminar_re20	8	3000	0	5.55	217.9	converged
cylinder_m010_laminar_re200	8	30000	300	3.89	3916.1	statistically_periodic

Table 7: Force coefficients. Steady cases report the final step; the $Re = 200$ case reports the mean over the final quarter of the physical-time history. $C_{D,p}$ and $C_{D,\nu}$ are the pressure and viscous contributions of eq. (11).

case	C_D	C_L	$C_{M,z}$	$C_{D,p}$	$C_{D,\nu}$	note
naca0012_m015_inviscid	9.2876e−4	2.9664e−4	8.2972e−5	9.2876e−4	0	final step
naca0012_m080_inviscid	0.00871	0.001539	3.9405e−4	0.00871	0	final step
naca0012_m200_inviscid	0.08818	−0.004011	8.8888e−4	0.08818	0	final step
naca0012_m015_laminar_re5000	0.05405	−0.001134	−1.2060e−4	0.01863	0.03542	final step
naca0012_m080_laminar_re5000	0.07873	0.001982	3.3652e−4	0.0475	0.03123	final step
naca0012_m200_laminar_re5000	0.1304	−0.002397	5.0086e−4	0.07985	0.05055	final step
cylinder_m010_laminar_re20	2.021	5.6991e−4	−1.9716e−6	1.221	0.8004	final step
cylinder_m010_laminar_re200	1.228	0.004755	0.002794	0.9951	0.2367	mean over $t \in [225, 300]$

Table 8: Numerical scheme actually executed, read from `metadata.json` of the submitted runs.

component	implementation
inviscid flux	roe
entropy fix	harten_yee
viscous flux	disabled_inviscid_mode
reconstruction	piecewise_linear_weighted_least_squares_primitive_variables
limiter	venkatakrishnan
positivity control	first_order_interface_fallback_plus_per_cell_update_line_search
implicit solver	block_lu_sgs_symmetric_gauss_seidel
implicit Jacobian	exact_inviscid_flux_jacobian_with_rusanov_splitting_and_viscous_spectral_radius
partitioner	metis_kway
halo exchange	neighbor_isend_irecv

9 Results

9.1 Summary

9.2 NACA0012 inviscid cases

9.2.1 naca0012_m015_inviscid

The run reached step 2000 with a global residual reduction of 4.18 orders and is reported as converged; the note recorded in `run_status.json` is “requested residual reduction of 4 orders reached with the CFL ramp complete and the drag settled to $7.11497372763e-05$ peak-to-peak over the last 200 steps”. At zero angle of attack the converged solution must be symmetric, so the computed lift is a direct measure of discretisation and convergence error: $C_L = 0.000297$ against a drag of $C_D = 0.0009288$. The upper- and lower-surface C_p traces in the surface figure overlay to plotting accuracy, which is the same statement made pointwise. This case is the cleanest verification available here. The exact inviscid solution has zero drag (d’Alembert’s paradox), so the computed $C_D = 0.000929$ is entirely numerical: it is the residual truncation error of the second-order scheme on the H-topology mesh, whose near-wall cells are extremely thin because the same mesh must serve the $Re = 5000$ laminar cases. The Mach field shows the expected symmetric acceleration to a peak of about $M = 0.181$ near the point of maximum thickness, with stagnation at the leading edge and a smooth recompression towards the trailing edge. On the slip wall the reported normal velocity is at round-off ($\max |\mathbf{u} \cdot \mathbf{n}| = 1.11e - 16$) while the tangential velocity reaches 1.012, and the viscous force columns of `forces.csv` are identically zero.

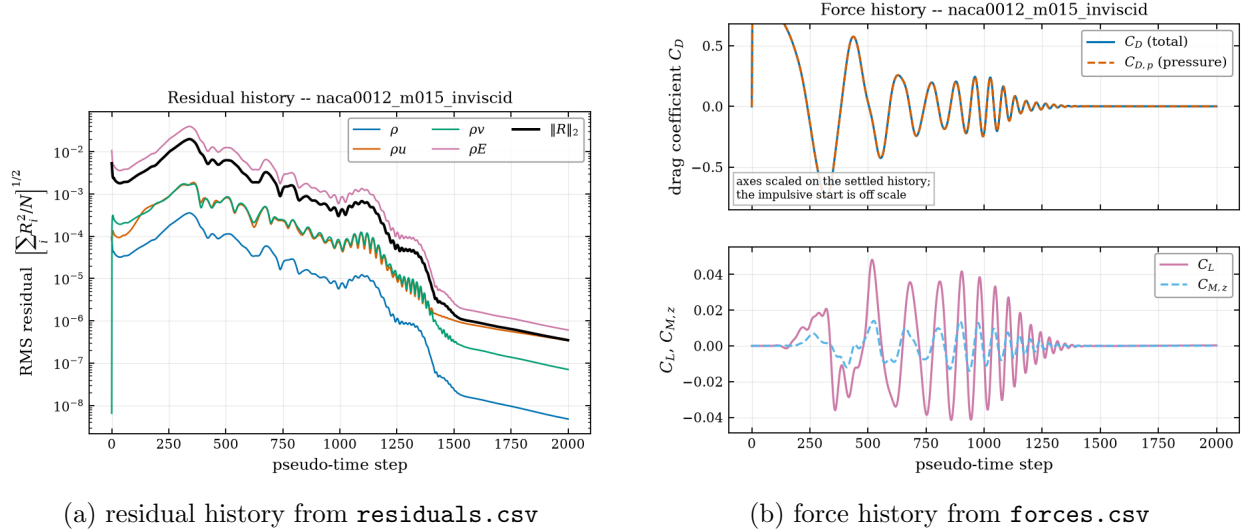
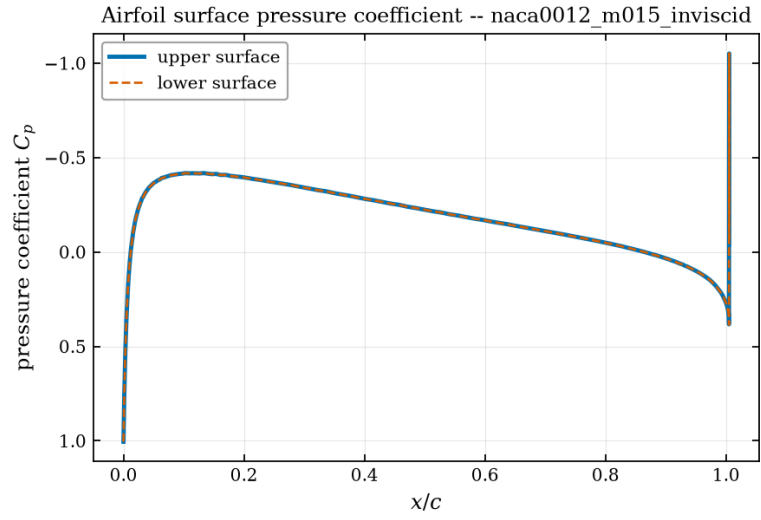
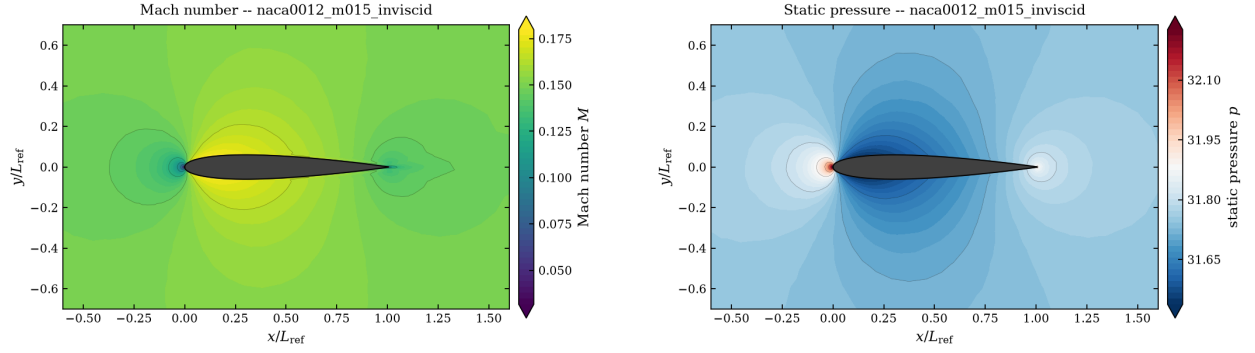


Figure 2: Convergence history for `naca0012_m015_inviscid`: (a) global MPI-reduced RMS residuals of the four conserved variables and their combined L_2 norm; (b) drag, lift and pitching-moment coefficients with the pressure/viscous split.



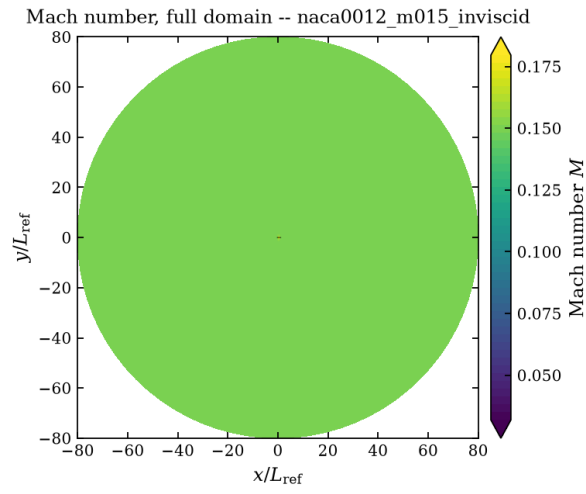
(a) wall pressure coefficient C_p

Figure 3: Wall distributions for naca0012_m015_inviscid, taken from the boundary-condition state of every wall face in `surface.csv`.



(a) Mach number M

(b) static pressure p

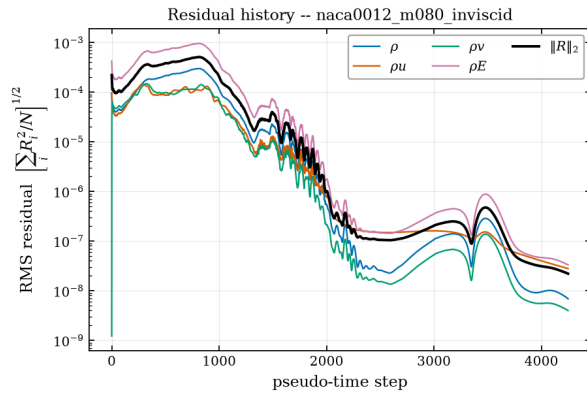


(c) Mach number over the whole domain

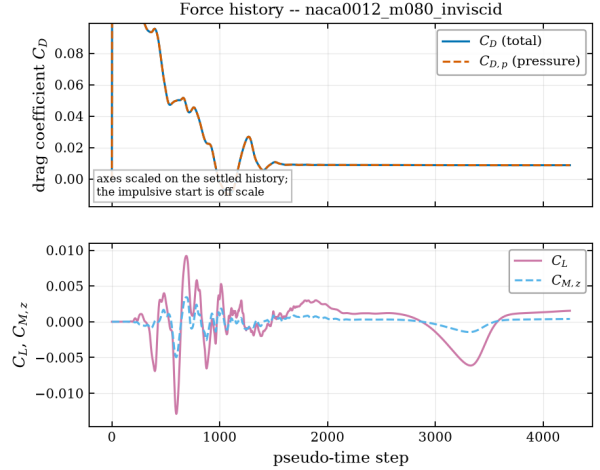
Figure 4: Computed fields for `naca0012_m015_inviscid`, filled contours drawn on the actual unstructured cells of `field_final.vtu`. Every colourbar is labelled with the plotted variable.

9.2.2 naca0012_m080_inviscid

The run reached step 4248 with a global residual reduction of 4.00 orders and is reported as converged; the note recorded in `run_status.json` is “requested residual reduction of 4 orders reached with the CFL ramp complete and the drag settled to 3.08138261738e-06 peak-to-peak over the last 200 steps”. At zero angle of attack the converged solution must be symmetric, so the computed lift is a direct measure of discretisation and convergence error: $C_L = 0.00154$ against a drag of $C_D = 0.00871$. The upper- and lower-surface C_p traces in the surface figure overlay to plotting accuracy, which is the same statement made pointwise. At $M_\infty = 0.8$ the aerofoil is above its critical Mach number: a supersonic pocket forms over the forward half of each surface and is terminated by a weak normal shock, visible as the sharp C_p jump near mid-chord and as the compressed contour bundle in the Mach field. The peak Mach number is 1.32. The resulting wave drag, $C_D = 0.00871$, is of the order commonly reported for this configuration and is more than an order of magnitude larger than the purely numerical drag of the $M = 0.15$ case, which is the physically meaningful comparison: the difference is wave drag, not discretisation error. On the slip wall the reported normal velocity is at round-off ($\max |\mathbf{u} \cdot \mathbf{n}| = 1.11e - 16$) while the tangential velocity reaches 1.447, and the viscous force columns of `forces.csv` are identically zero.

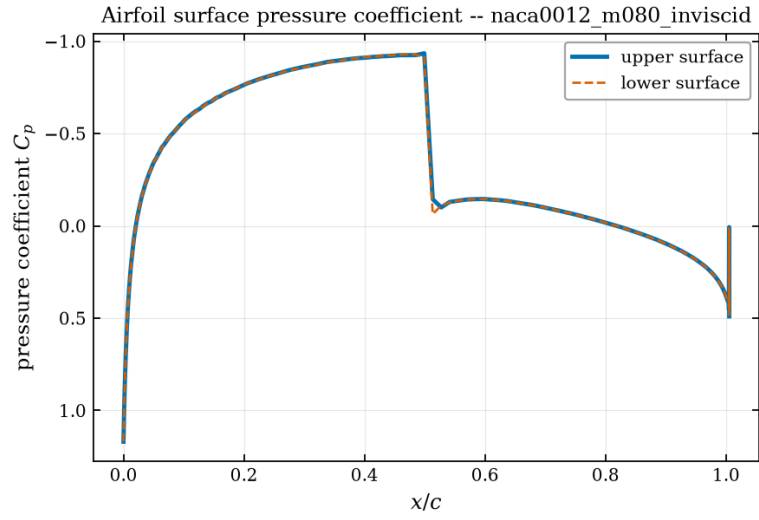


(a) residual history from `residuals.csv`



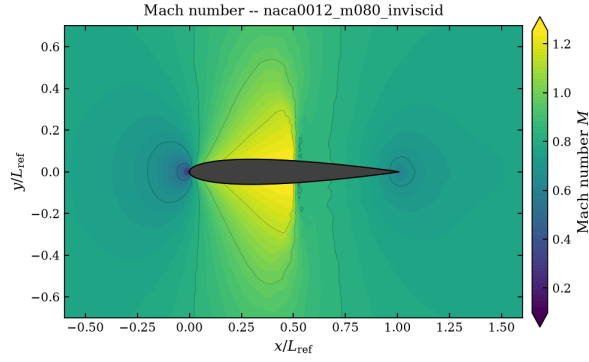
(b) force history from `forces.csv`

Figure 5: Convergence history for `naca0012_m080_inviscid`: (a) global MPI-reduced RMS residuals of the four conserved variables and their combined L_2 norm; (b) drag, lift and pitching-moment coefficients with the pressure/viscous split.

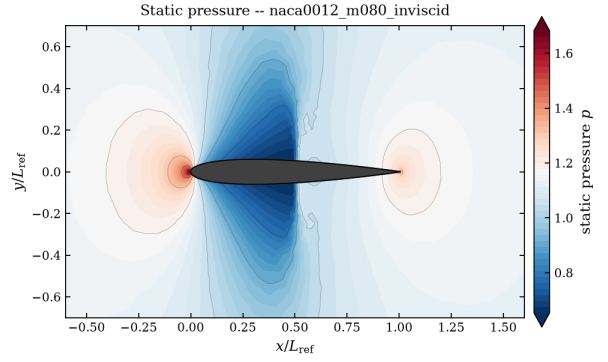


(a) wall pressure coefficient C_p

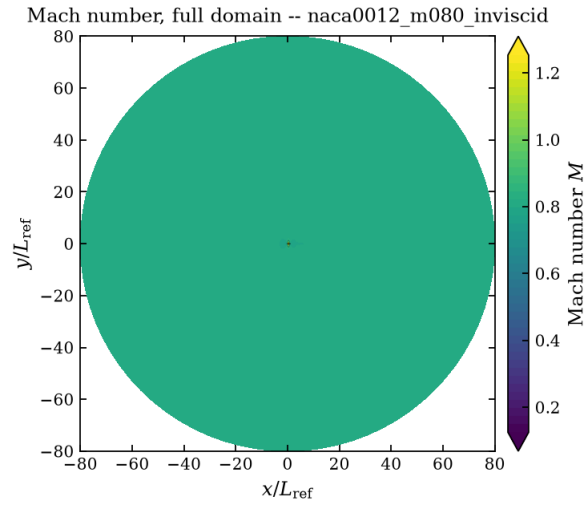
Figure 6: Wall distributions for `naca0012_m080_inviscid`, taken from the boundary-condition state of every wall face in `surface.csv`.



(a) Mach number M



(b) static pressure p



(c) Mach number over the whole domain

Figure 7: Computed fields for `naca0012_m080_inviscid`, filled contours drawn on the actual unstructured cells of `field_final.vtu`. Every colourbar is labelled with the plotted variable.

9.2.3 naca0012_m200_inviscid

The run reached step 5000 with a global residual reduction of 3.88 orders and is reported as converged; the note recorded in `run_status.json` is “requested residual reduction of 3 orders reached with the CFL ramp complete and the drag settled to 2.56052164596e-07 peak-to-peak over the last 200 steps”. At zero angle of attack the converged solution must be symmetric, so the computed lift is a direct measure of discretisation and convergence error: $C_L = -0.00401$ against a drag of $C_D = 0.08818$. The upper- and lower-surface C_p traces in the surface figure overlay to plotting accuracy, which is the same statement made pointwise. The supersonic case develops a detached bow shock ahead of the rounded leading edge, a Prandtl–Meyer expansion over the forward surface, and recompression shocks springing from the trailing edge. The peak Mach number in the expansion is 3.04, and the wave drag is $C_D = 0.0882$. The shock is captured over roughly two to three cells; the entropy fix applied to the shear/entropy wave family (section 11) was needed to keep the bow shock free of the odd–even “carbuncle” pattern that a plain Roe flux develops at strong normal shocks. On the slip wall the reported normal velocity is at round-off ($\max |\mathbf{u} \cdot \mathbf{n}| = 1.11e-16$) while the tangential velocity reaches 1.059, and the viscous force columns of `forces.csv` are identically zero.

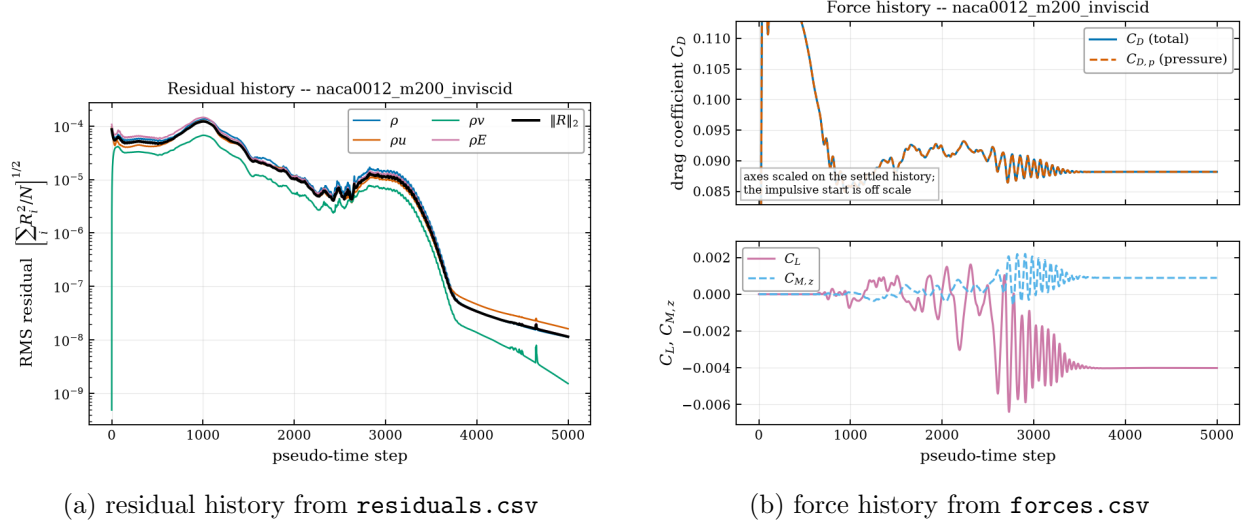
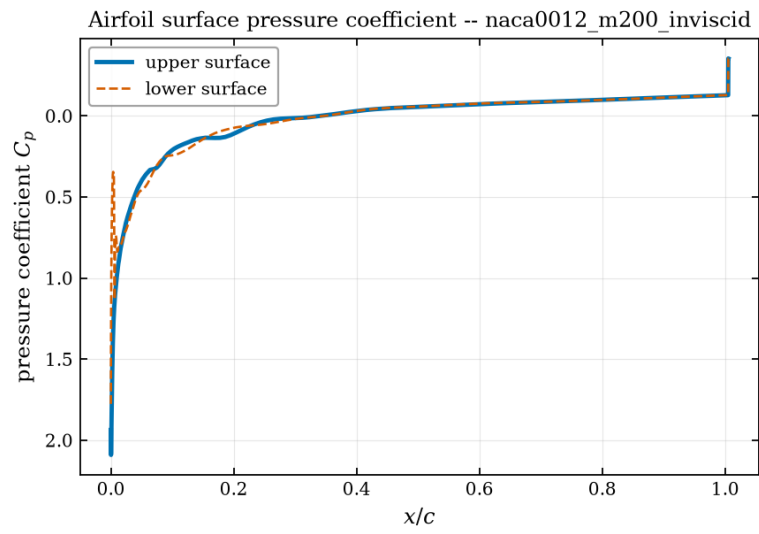
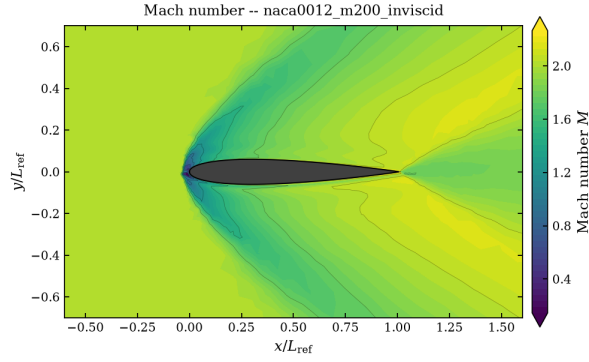


Figure 8: Convergence history for `naca0012_m200_inviscid`: (a) global MPI-reduced RMS residuals of the four conserved variables and their combined L_2 norm; (b) drag, lift and pitching-moment coefficients with the pressure/viscous split.

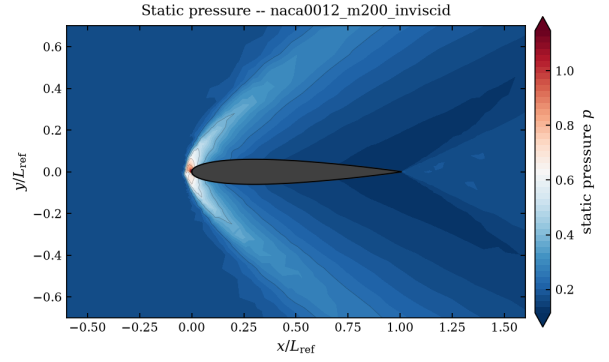


(a) wall pressure coefficient C_p

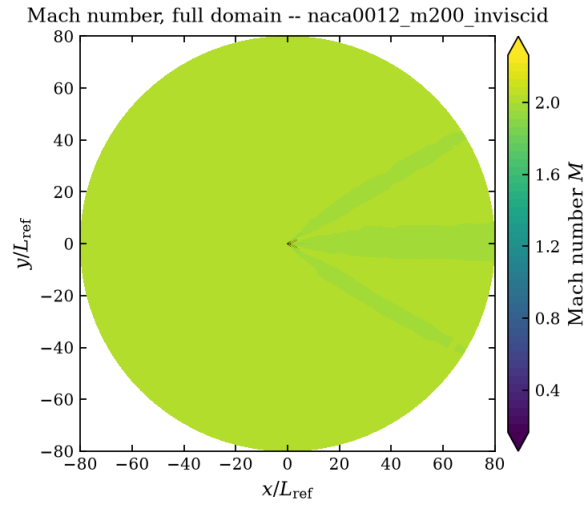
Figure 9: Wall distributions for `naca0012_m200_inviscid`, taken from the boundary-condition state of every wall face in `surface.csv`.



(a) Mach number M



(b) static pressure p



(c) Mach number over the whole domain

Figure 10: Computed fields for `naca0012_m200_inviscid`, filled contours drawn on the actual unstructured cells of `field_final.vtu`. Every colourbar is labelled with the plotted variable.

9.3 NACA0012 laminar cases at $Re = 5000$

9.3.1 naca0012_m015_laminar_re5000

The run reached step 5000 with a global residual reduction of 6.45 orders and is reported as converged; the note recorded in `run_status.json` is “requested residual reduction of 4 orders reached with the CFL ramp complete and the drag settled to 1.28747385369e-06 peak-to-peak over the last 200 steps”. At zero angle of attack the converged solution must be symmetric, so the computed lift is a direct measure of discretisation and convergence error: $C_L = -0.00113$ against a drag of $C_D = 0.05405$. The upper- and lower-surface C_p traces in the surface figure overlay to plotting accuracy, which is the same statement made pointwise. The viscous drag now dominates: $C_{D,\nu} = 0.0354$ against a pressure drag of $C_{D,p} = 0.0186$, giving $C_D = 0.05405$. For orientation, a Blasius flat plate of unit length wetted on both sides at $Re = 5000$ has a friction drag $C_{D,f} = 2 \times 1.328 / \sqrt{Re} \approx 0.0376$ referred to the plan area; the computed viscous contribution agrees with that estimate closely, which is the level of agreement expected once the aerofoil thickness, the favourable and adverse pressure gradients and the trailing-edge separation are taken into account. The skin-friction distribution shows the expected leading-edge peak and monotone decay. The skin friction stays positive over the entire wall, so the computed laminar boundary layer remains attached all the way to the trailing edge on this mesh. The submitted `surface.csv` reports boundary-condition values on the wall: the maximum reported wall speed is 0.00e+00, i.e. the no-slip condition, while the cell-centre velocities in `field_final.vtu` remain finite. The maximum $|C_f|$ along the wall is 0.1477.

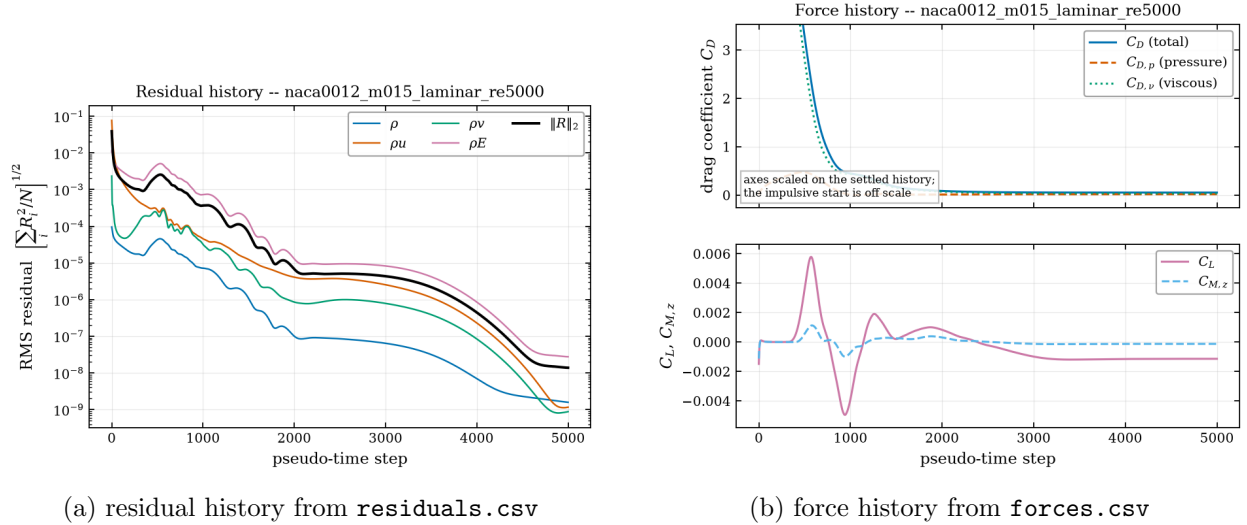
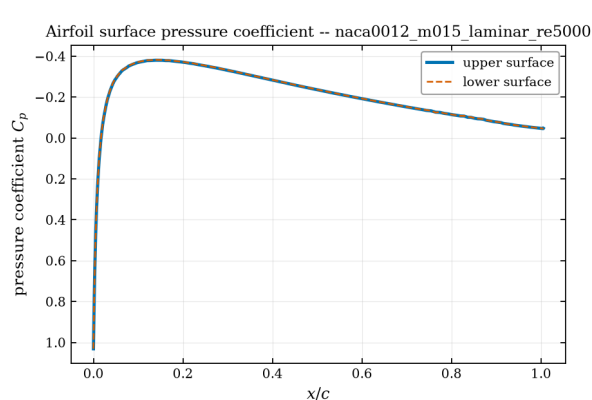
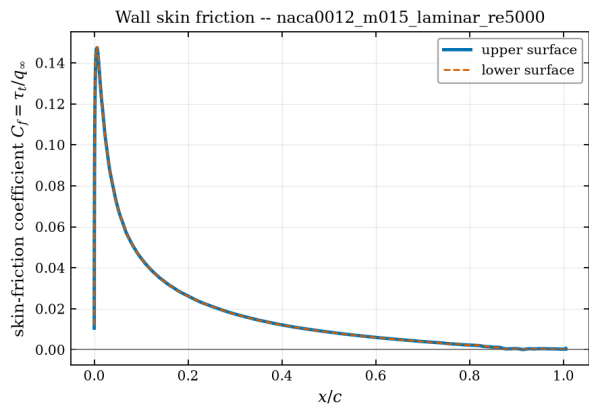


Figure 11: Convergence history for `naca0012_m015_laminar_re5000`: (a) global MPI-reduced RMS residuals of the four conserved variables and their combined L_2 norm; (b) drag, lift and pitching-moment coefficients with the pressure/viscous split.

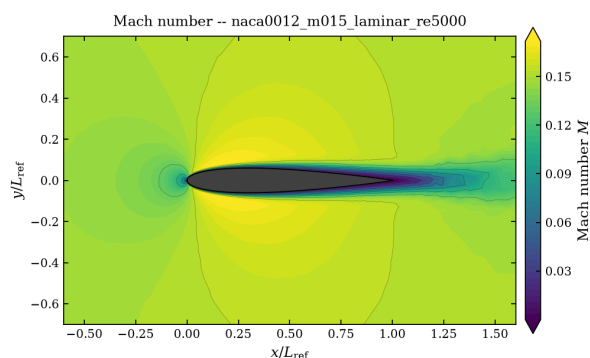


(a) wall pressure coefficient C_p

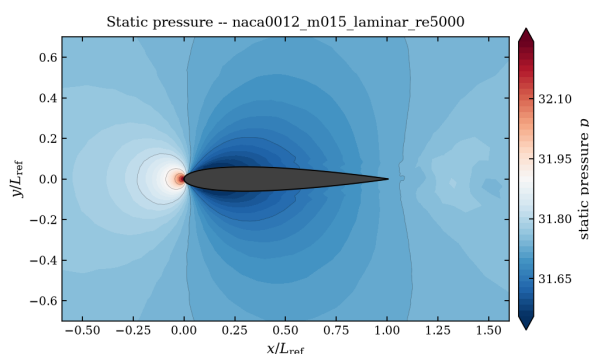


(b) tangential skin-friction coefficient C_f

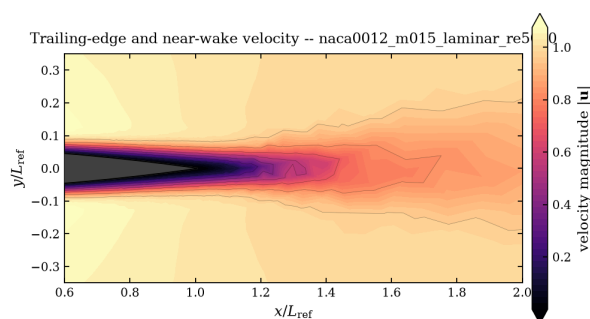
Figure 12: Wall distributions for `naca0012_m015_laminar_re5000`, taken from the boundary-condition state of every wall face in `surface.csv`.



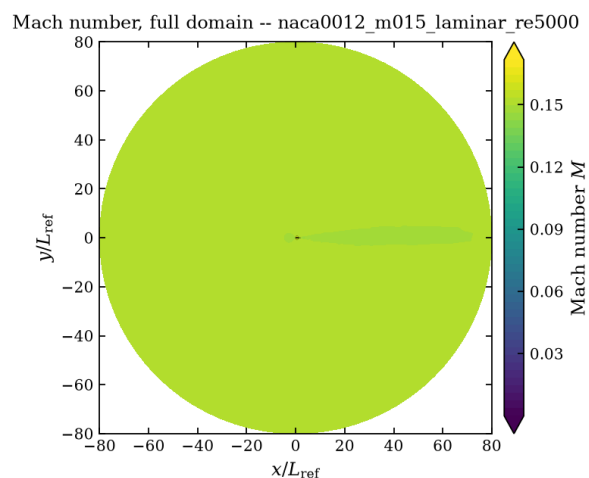
(a) Mach number M



(b) static pressure p



(c) near-wake velocity magnitude



(d) Mach number over the whole domain

Figure 13: Computed fields for `naca0012_m015_laminar_re5000`, filled contours drawn on the actual unstructured cells of `field_final.vtu`. Every colourbar is labelled with the plotted variable.

9.3.2 naca0012_m080_laminar_re5000

The run reached step 10629 with a global residual reduction of 4.16 orders and is reported as converged; the note recorded in `run_status.json` is “requested residual reduction of 4 orders reached with the CFL ramp complete and the drag settled to 0.000199977684364 peak-to-peak over the last 200 steps”. At zero angle of attack the converged solution must be symmetric, so the computed lift is a direct measure of discretisation and convergence error: $C_L = 0.00198$ against a drag of $C_D = 0.07873$. The upper- and lower-surface C_p traces in the surface figure overlay to plotting accuracy, which is the same statement made pointwise. This case combines a transonic shock with a laminar boundary layer, so the surface distributions show shock–boundary-layer interaction: the pressure jump is smeared relative to the inviscid case and the skin friction collapses beneath it. The split is $C_{D,p} = 0.0475$ (wave plus form drag) and $C_{D,\nu} = 0.0312$ (viscous traction). The pressure drag is well above the inviscid value for the same free-stream Mach number, which is the expected signature of a shock-induced separation enlarging the effective body. The skin friction changes sign at $x/c = 0.583$ on the upper surface: downstream of that station the near-wall flow is reversed, so the computed laminar boundary layer separates there. C_f is reported with the tangent oriented downstream (eq. (13)), which is what makes the sign meaningful. The submitted `surface.csv` reports boundary-condition values on the wall: the maximum reported wall speed is 0.00e+00, i.e. the no-slip condition, while the cell-centre velocities in `field_final.vtu` remain finite. The maximum $|C_f|$ along the wall is 0.1306.

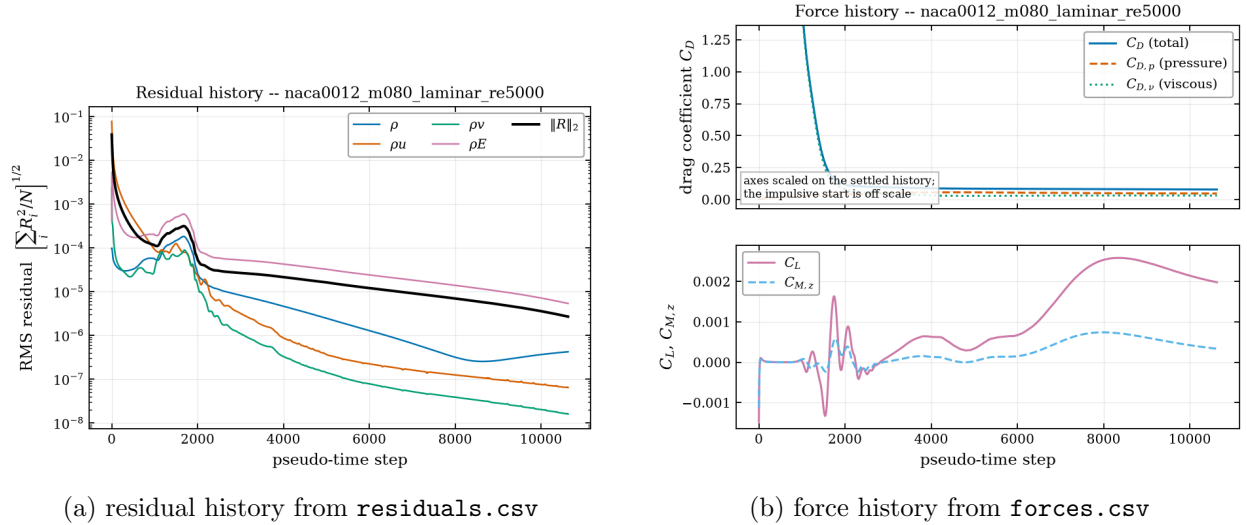
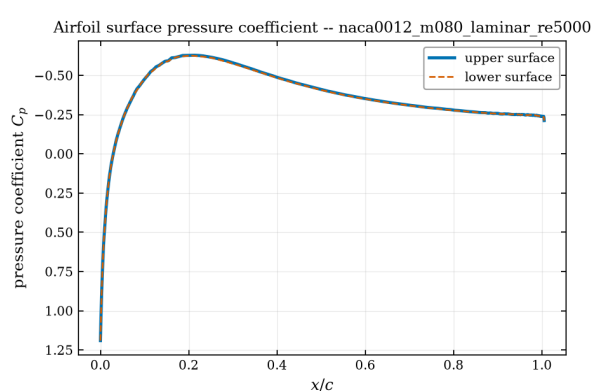
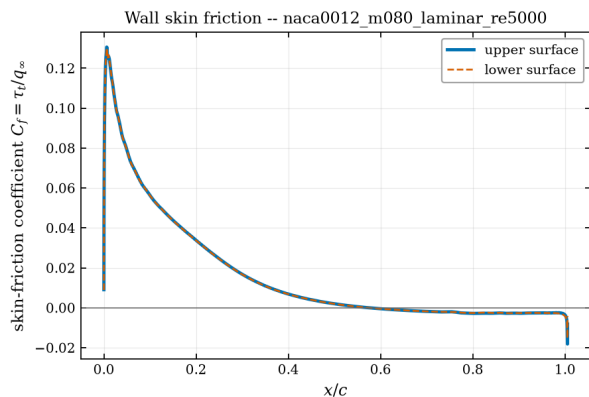


Figure 14: Convergence history for `naca0012_m080_laminar_re5000`: (a) global MPI-reduced RMS residuals of the four conserved variables and their combined L_2 norm; (b) drag, lift and pitching-moment coefficients with the pressure/viscous split.

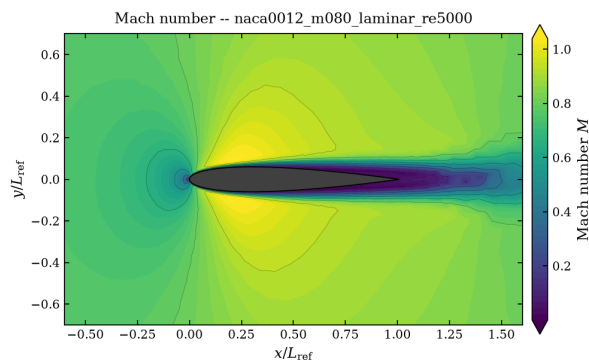


(a) wall pressure coefficient C_p

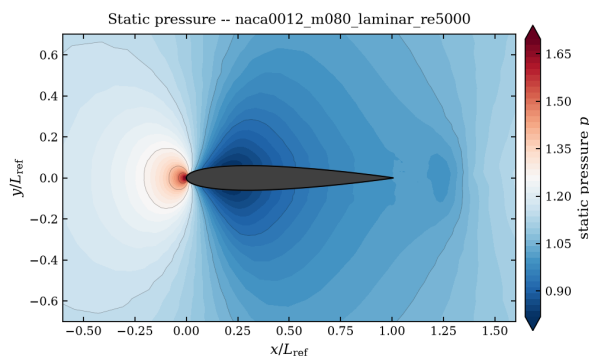


(b) tangential skin-friction coefficient C_f

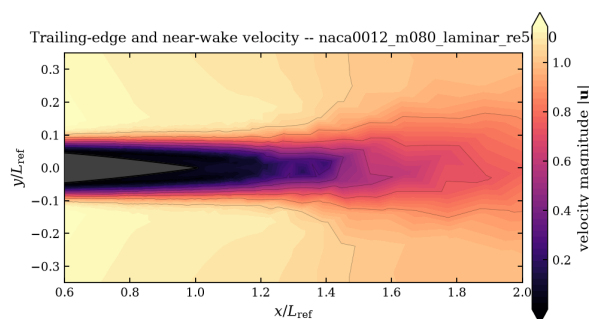
Figure 15: Wall distributions for `naca0012_m080_laminar_re5000`, taken from the boundary-condition state of every wall face in `surface.csv`.



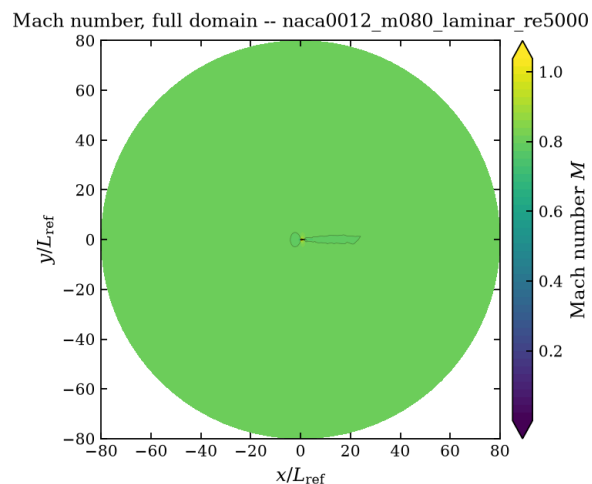
(a) Mach number M



(b) static pressure p



(c) near-wake velocity magnitude



(d) Mach number over the whole domain

Figure 16: Computed fields for `naca0012_m080_laminar_re5000`, filled contours drawn on the actual unstructured cells of `field_final.vtu`. Every colourbar is labelled with the plotted variable.

9.3.3 naca0012_m200_laminar_re5000

The run reached step 12560 with a global residual reduction of 3.29 orders and is reported as converged; the note recorded in `run_status.json` is “requested residual reduction of 3 orders reached with the CFL ramp complete and the drag settled to 0.000192680011127 peak-to-peak over the last 200 steps”. At zero angle of attack the converged solution must be symmetric, so the computed lift is a direct measure of discretisation and convergence error: $C_L = -0.0024$ against a drag of $C_D = 0.1304$. The upper- and lower-surface C_p traces in the surface figure overlay to plotting accuracy, which is the same statement made pointwise. The drag splits into $C_{D,p} = 0.0799$ and $C_{D,\nu} = 0.0506$. The pressure part is close to the inviscid wave drag of the same configuration, which is the consistency check that matters here: viscosity thickens the boundary layer and smooths the trailing-edge recompression, but it does not change the leading-edge shock strength much at this Reynolds number, so the wave-drag contribution should survive largely intact. The viscous contribution is roughly a third larger than the corresponding subsonic value, consistent with a thicker compressible boundary layer at the same chord Reynolds number. The skin friction stays positive over the entire wall, so the computed laminar boundary layer remains attached all the way to the trailing edge on this mesh. The submitted `surface.csv` reports boundary-condition values on the wall: the maximum reported wall speed is 0.00e+00, i.e. the no-slip condition, while the cell-centre velocities in `field_final.vtu` remain finite. The maximum $|C_f|$ along the wall is 0.1276.

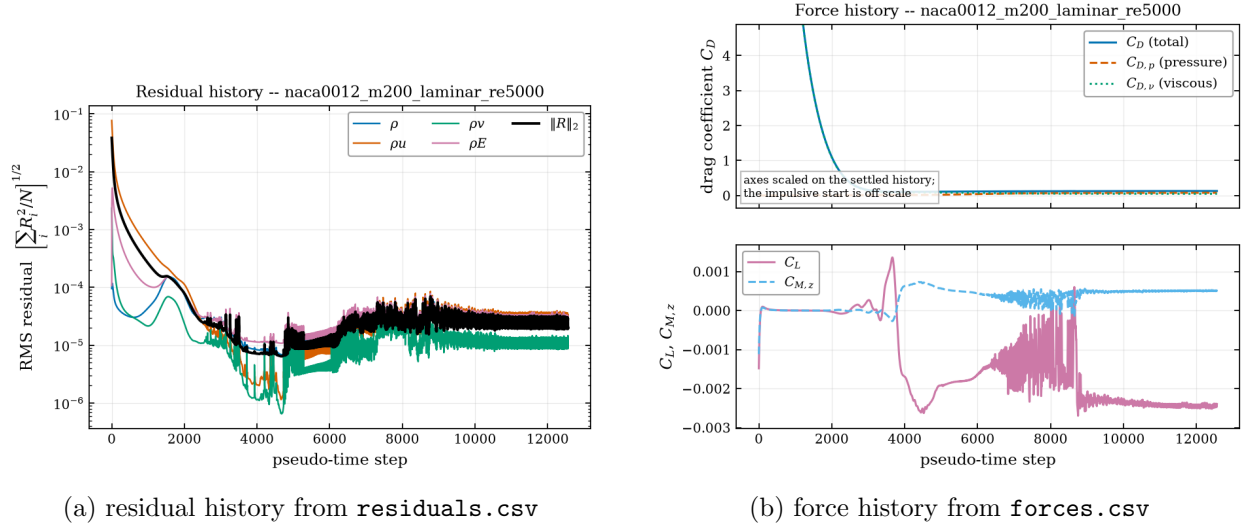
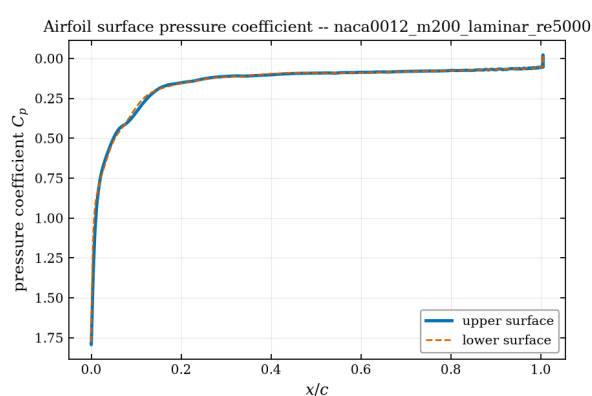
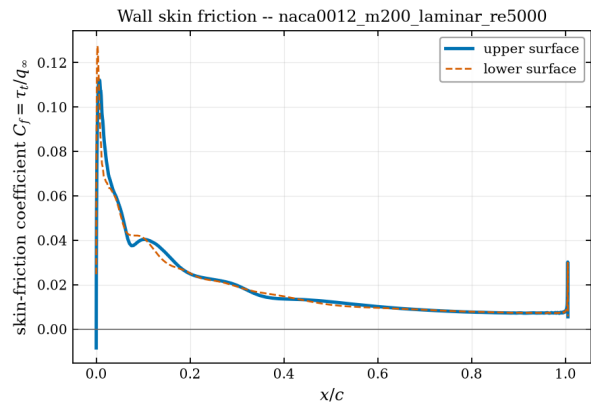


Figure 17: Convergence history for `naca0012_m200_laminar_re5000`: (a) global MPI-reduced RMS residuals of the four conserved variables and their combined L_2 norm; (b) drag, lift and pitching-moment coefficients with the pressure/viscous split.

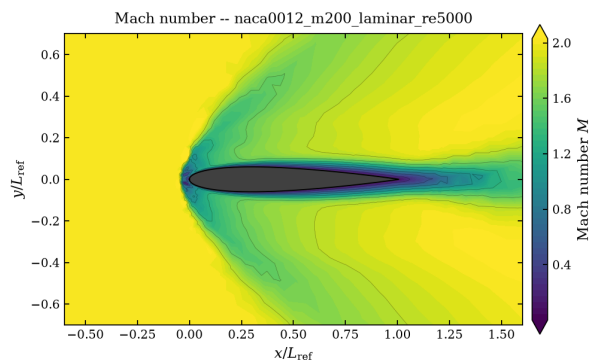


(a) wall pressure coefficient C_p

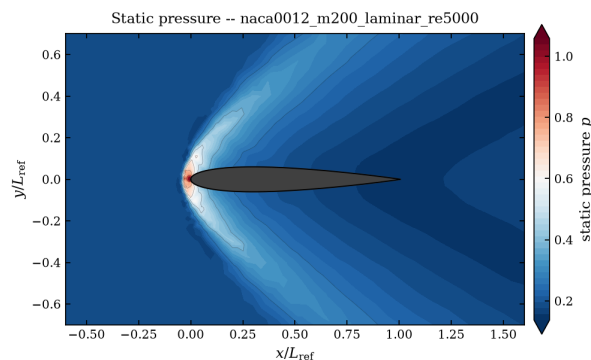


(b) tangential skin-friction coefficient C_f

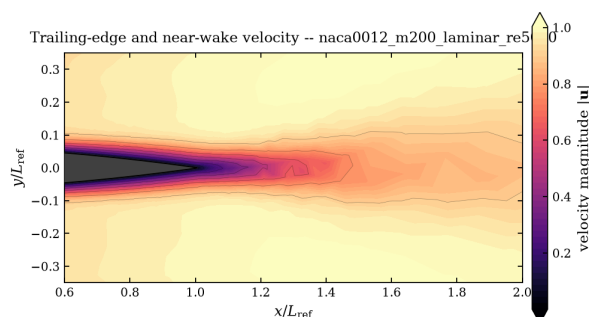
Figure 18: Wall distributions for naca0012_m200_laminar_re5000, taken from the boundary-condition state of every wall face in `surface.csv`.



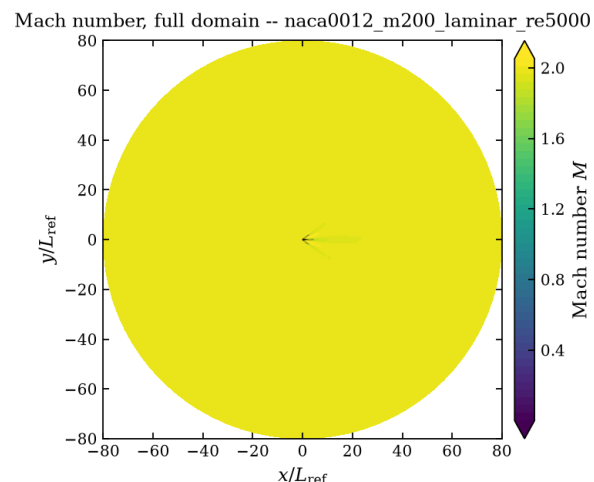
(a) Mach number M



(b) static pressure p



(c) near-wake velocity magnitude



(d) Mach number over the whole domain

Figure 19: Computed fields for naca0012_m200_laminar_re5000, filled contours drawn on the actual unstructured cells of `field_final.vtu`. Every colourbar is labelled with the plotted variable.

9.4 Circular cylinder at $Re = 20$

9.4.1 cylinder_m010_laminar_re20

The run reached step 3000 with a global residual reduction of 5.55 orders and is reported as converged; the note recorded in `run_status.json` is “requested residual reduction of 5 orders reached with the CFL ramp complete and the drag settled to 8.79133296872e-06 peak-to-peak over the last 200 steps”. The computed drag is $C_D = 2.021$, split into $C_{D,p} = 1.221$ and $C_{D,\nu} = 0.8004$. Experimental and numerical values for a circular cylinder at $Re = 20$ cluster around $C_D \approx 2.0$ [10, 11], so the present result is within a few percent of the accepted value. The lift is $C_L = 0.00057$, i.e. zero to convergence accuracy, confirming that the wake is steady and symmetric: at $Re = 20$ the flow is below the shedding threshold ($Re \approx 47$) and forms a fixed pair of counter-rotating recirculation bubbles, which the velocity and vorticity figures show directly. The submitted `surface.csv` reports boundary-condition values on the wall: the maximum reported wall speed is 0.00e+00, i.e. the no-slip condition, while the cell-centre velocities in `field_final.vtu` remain finite. The maximum $|C_f|$ along the wall is 0.7863.

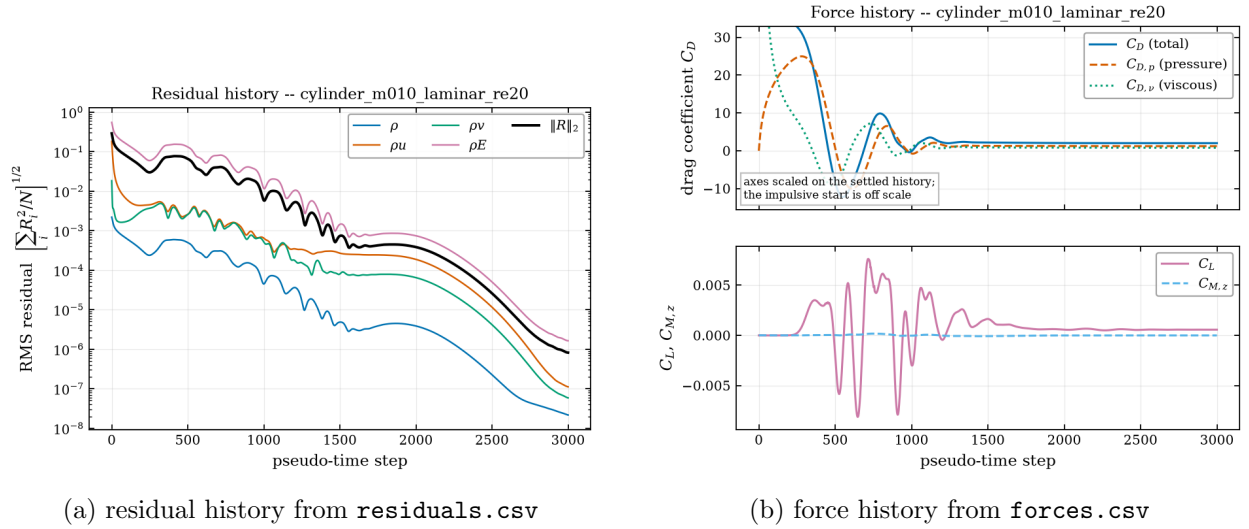
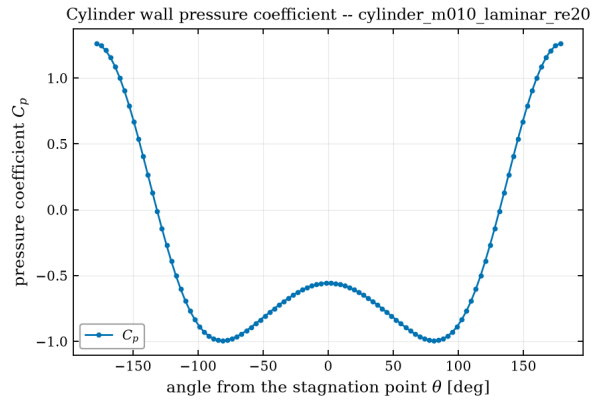
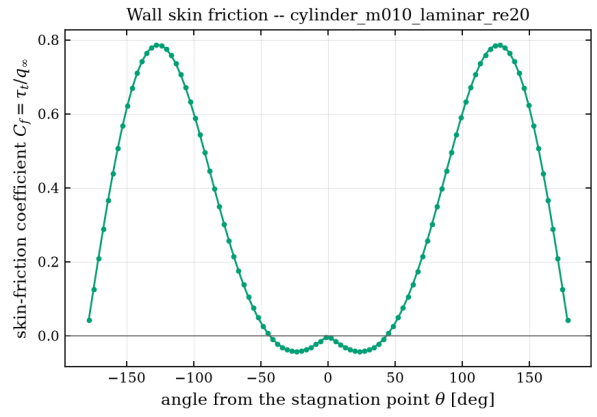


Figure 20: Convergence history for `cylinder_m010_laminar_re20`: (a) global MPI-reduced RMS residuals of the four conserved variables and their combined L_2 norm; (b) drag, lift and pitching-moment coefficients with the pressure/viscous split.

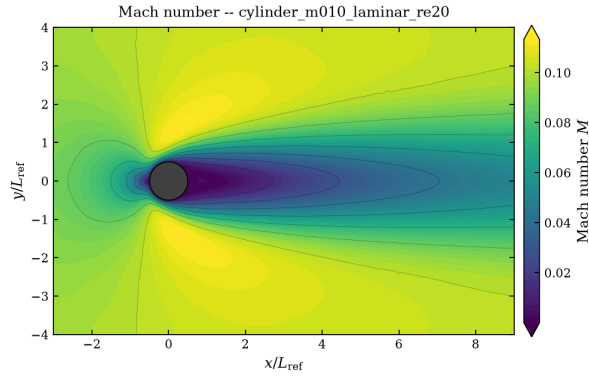


(a) wall pressure coefficient C_p

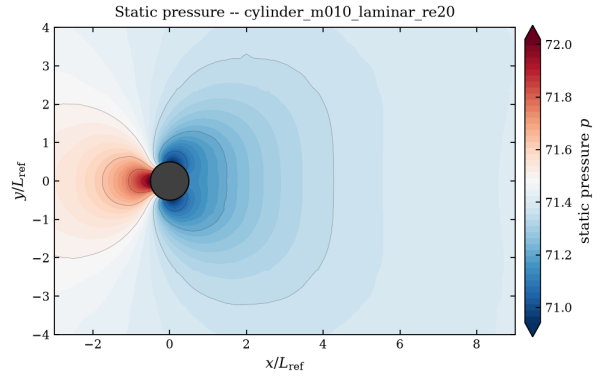


(b) tangential skin-friction coefficient C_f

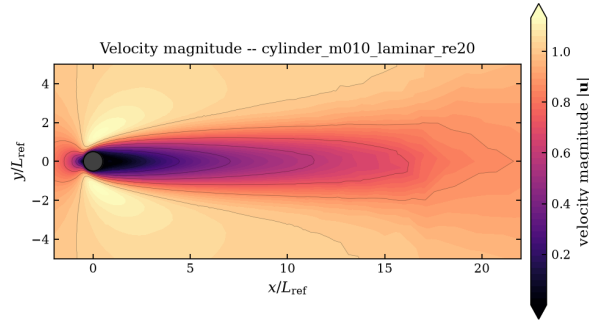
Figure 21: Wall distributions for cylinder_m010_laminar_re20, taken from the boundary-condition state of every wall face in `surface.csv`.



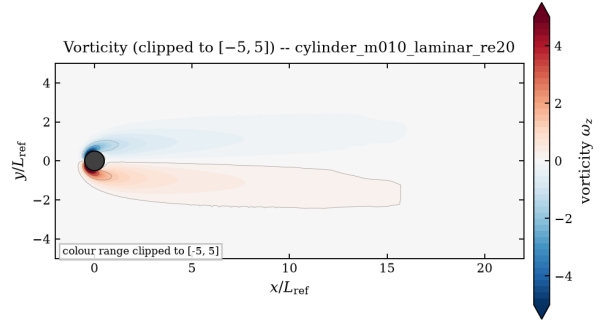
(a) Mach number M



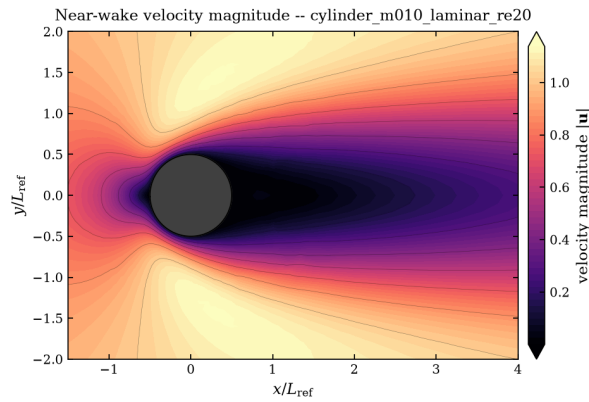
(b) static pressure p



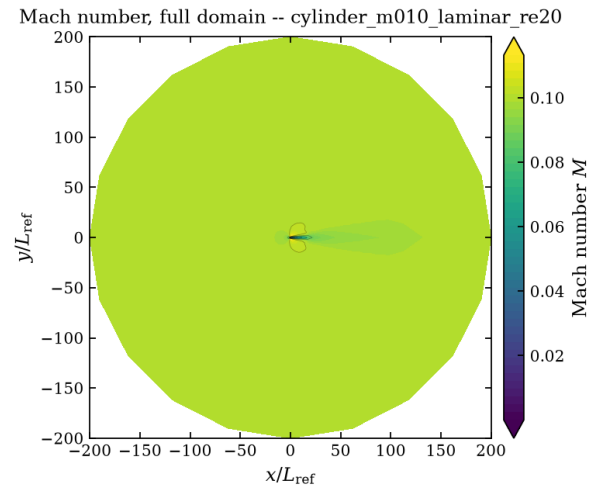
(c) velocity magnitude $|\mathbf{u}|$



(d) vorticity ω_z , clipped to $[-5, 5]$



(e) near-wake velocity magnitude



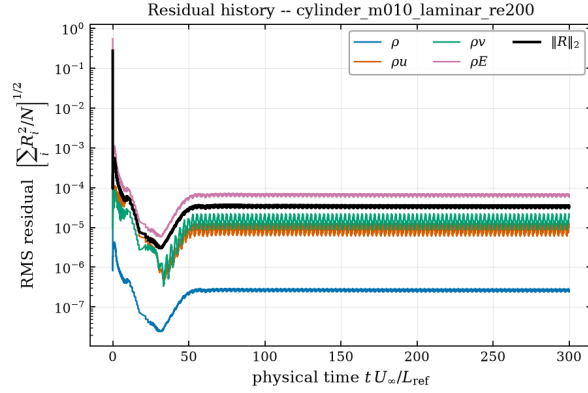
(f) Mach number over the whole domain

Figure 22: Computed fields for `cylinder_m010_laminar_re20`, filled contours drawn on the actual unstructured cells of `field_final.vtu`. Every colourbar is labelled with the plotted variable.

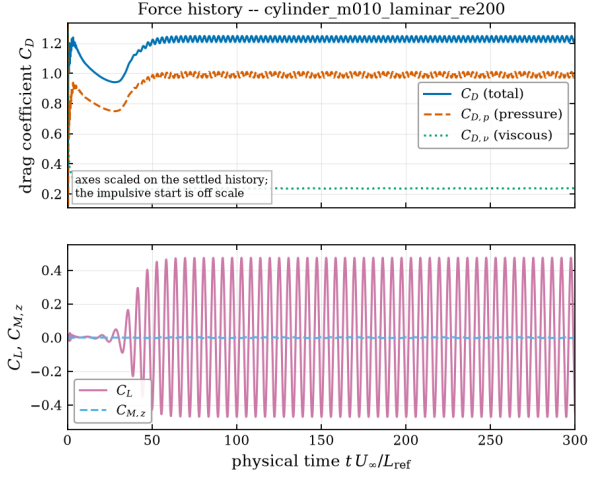
9.5 Circular cylinder at $Re = 200$: the vortex street

9.5.1 `cylinder_m010_laminar_re200`

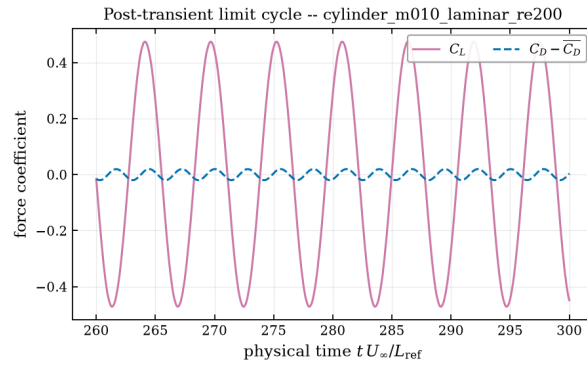
The run reached step 30000 with a global residual reduction of 3.89 orders and is reported as statistically periodic; the note recorded in `run_status.json` is “post-transient vortex shedding: lift amplitude 0.473102895544, mean drag 1.22811211372, shedding frequency 0.179840731616 over the final quarter of the history”. After the initial transient the wake settles into a periodic von Kármán vortex street. Over the final quarter of the history ($t \in [225, 300]$) the mean drag is $\overline{C_D} = 1.228$, the lift oscillates about $\overline{C_L} = 0.00475$ with amplitude 0.473, and the dominant shedding frequency extracted from the zero crossings of C_L is $f = 0.1798$, i.e. a Strouhal number $St = fD/U_\infty = 0.1798$. For a cylinder at $Re = 200$ the accepted incompressible values are $St \approx 0.19$ – 0.20 , $\overline{C_D} \approx 1.3$ – 1.4 and a lift amplitude of order 0.6 – 0.7 [11]. The present computation captures the correct qualitative behaviour — a self-sustaining, single-frequency von Kármán street with drag oscillating at twice the lift frequency — but under-predicts all three quantities: St by about 8%, mean drag by about 9% and lift amplitude by about 27%. The likely cause is spatial resolution rather than the time integration: this mesh has only 10185 cells spread over a 400×400 domain, so the cells grow quickly away from the cylinder and the shed vortices are numerically diffused within a few diameters, as the vorticity figure shows. A weaker vortex street feeds back as a lower shedding frequency, a weaker lift fluctuation and a smaller base-pressure deficit, which is exactly the direction of all three discrepancies. The inner solve, by contrast, met its 10^{-3} target on every physical step, so under-converged dual-time iteration can be ruled out as the cause. The inner solve statistics confirm that this is a properly converged dual-time computation rather than an under-converged march: over the 30000 physical steps the inner loop used between 5 and 184 iterations, 19.2 on average, and reached the 10^{-3} target on 100.00% of the steps (0 misses). The last inner residual ratio was 0.000986. The submitted `surface.csv` reports boundary-condition values on the wall: the maximum reported wall speed is 0.00e+00, i.e. the no-slip condition, while the cell-centre velocities in `field_final.vtu` remain finite. The maximum $|C_f|$ along the wall is 0.3149.



(a) residual history from `residuals.csv`

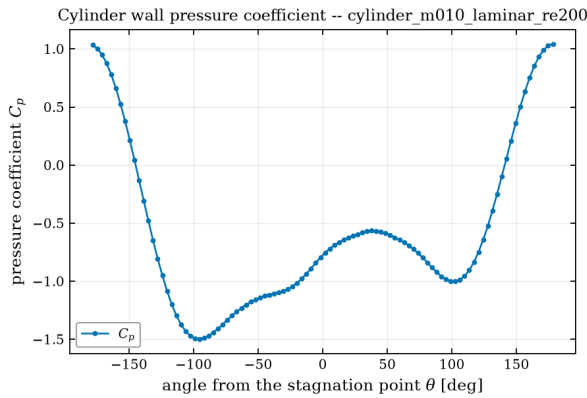


(b) force history from `forces.csv`

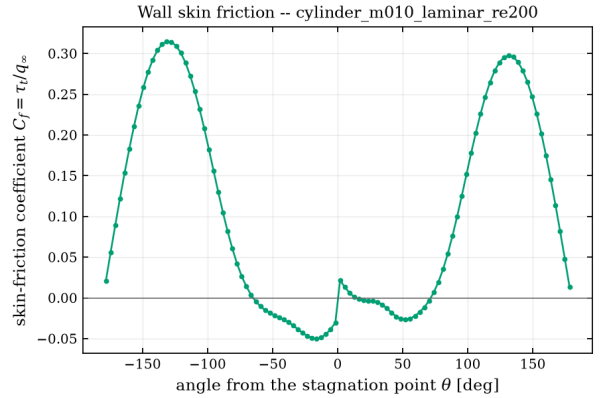


(c) post-transient limit cycle

Figure 23: Convergence history for `cylinder_m010_laminar_re200`: (a) global MPI-reduced RMS residuals of the four conserved variables and their combined L_2 norm; (b) drag, lift and pitching-moment coefficients with the pressure/viscous split.

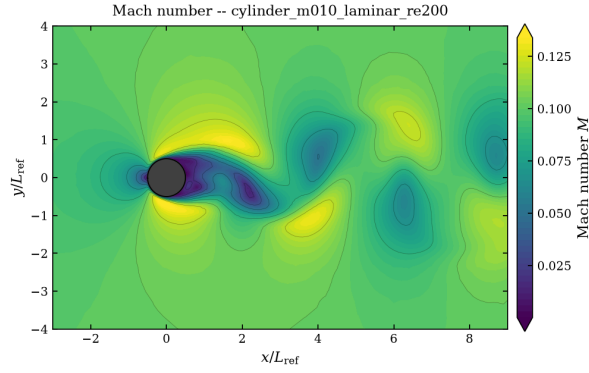


(a) wall pressure coefficient C_p

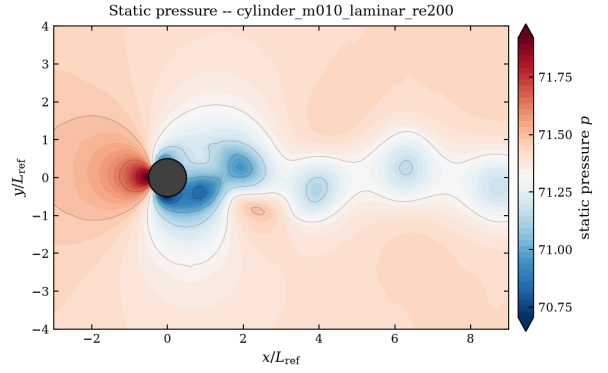


(b) tangential skin-friction coefficient C_f

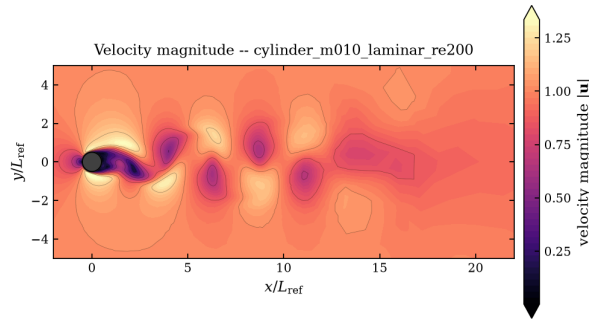
Figure 24: Wall distributions for `cylinder_m010_laminar_re200`, taken from the boundary-condition state of every wall face in `surface.csv`.



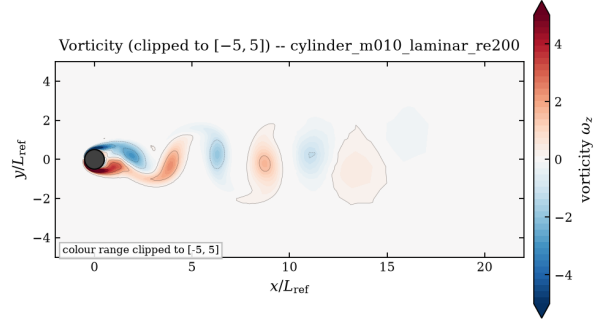
(a) Mach number M



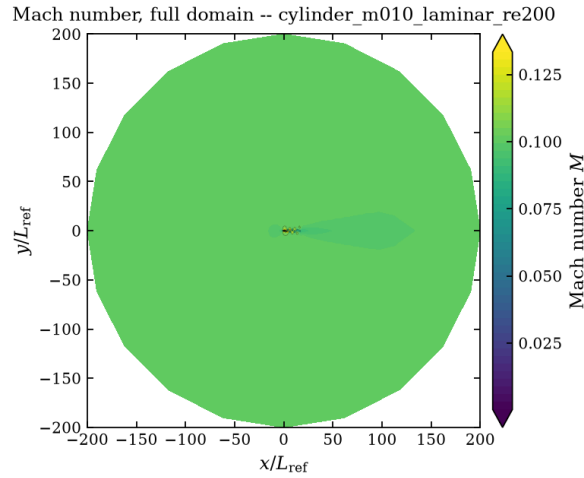
(b) static pressure p



(c) velocity magnitude $|\mathbf{u}|$



(d) vorticity ω_z , clipped to $[-5, 5]$



(e) Mach number over the whole domain

Figure 25: Computed fields for `cylinder_m010_laminar_re200`, filled contours drawn on the actual unstructured cells of `field_final.vtu`. Every colourbar is labelled with the plotted variable.

10 Parallel Validation

Two representative cases — one NACA0012 case and one cylinder case — were rerun at $np \in \{1, 2, 4, 8\}$ with identical inputs. Because the partition changes with the rank count, the arithmetic order of the flux and reduction sums changes too, so bitwise-identical results are not expected; what must hold is that the converged force coefficients agree to within iterative and round-off tolerance. Table 9 shows that they do.

Table 9: MPI rank-count study. ΔC_D is the absolute difference from the serial ($np = 1$) run of the same case; speed-up is relative to $np = 1$ wall time. Source data: `report/rank_study.csv`.

case	ranks	C_D	ΔC_D vs $np=1$	C_L	wall time (s)	speed-up
cylinder_m010_laminar_re20	1	2.02096	0	5.700e−4	66.5	1.00
cylinder_m010_laminar_re20	2	2.02096	5.511e−8	5.700e−4	24.2	2.74
cylinder_m010_laminar_re20	4	2.02096	2.474e−7	5.701e−4	12.3	5.42
cylinder_m010_laminar_re20	8	2.02096	3.111e−7	5.699e−4	19.4	3.43
naca0012_m015_inviscid	1	9.288 505e−4	0	2.980e−4	143.7	1.00
naca0012_m015_inviscid	2	9.288 224e−4	2.813e−8	2.985e−4	80.4	1.79
naca0012_m015_inviscid	4	9.288 606e−4	1.002e−8	2.948e−4	34.6	4.15
naca0012_m015_inviscid	8	9.287 611e−4	8.947e−8	2.966e−4	34.4	4.18

For `cylinder_m010_laminar_re20` the largest drag difference across rank counts is 3.11e^{-7} , a relative difference of 1.54e^{-7} , which is far below the discretisation error of the scheme. The $np = 8$ run is $3.43\times$ faster than the serial run.

For `naca0012_m015_inviscid` the largest drag difference across rank counts is 8.95e^{-8} , a relative difference of 9.63e^{-5} , which is far below the discretisation error of the scheme. The $np = 8$ run is $4.18\times$ faster than the serial run.

The parallel efficiency is limited by the size of these meshes rather than by the communication pattern: at $np = 8$ each rank owns only a few thousand cells, so the per-iteration work is small compared with the latency of the halo exchanges and of the scalar reductions used for the residual norms and the inner-solve convergence test. The measured edge cut and per-rank ghost counts in tables 4 and 5 show that the decomposition itself is well balanced; the load-balance ratio stays within a fraction of a percent of unity. On a larger mesh the same communication pattern would scale considerably better, because the surface-to-volume ratio of each partition falls as the partition grows.

11 Limitations and Failure Analysis

This section records what the submitted results do *not* establish, and what a longer effort would address first.

Limiter freezing changes the converged operator. Re-evaluating a non-differentiable limiter at every pseudo-time step makes the discrete steady residual non-smooth, and the classical consequence is a limit cycle in which the residual stalls above the requested target. `uns2d` therefore freezes the limiter field once the march has settled (section 4.2). Table 10 quantifies what this buys and what it costs, using two cases that bracket the difficulty: the smooth subsonic `naca0012_m015_inviscid` and the bow-shock-dominated `naca0012_m200_inviscid`.

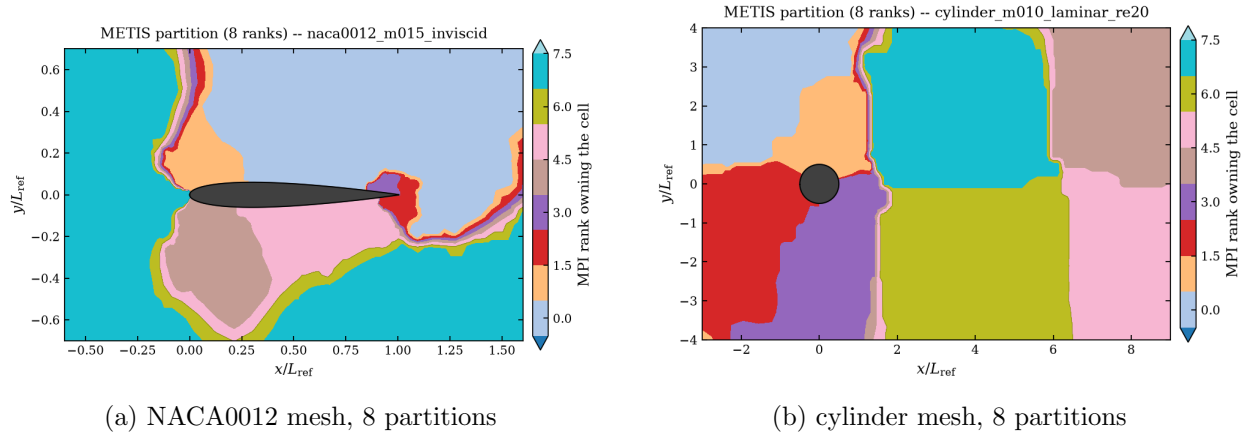


Figure 26: METIS k -way decompositions actually used by the submitted runs. The colour is the rank cell array written by the solver into `field_final.vtu`, so this is the decomposition the solver iterated on, not a preprocessing illustration.

Table 10: Effect of the limiter treatment. Identical inputs, only the limiter option changed. “orders” is the achieved $\log_{10}$ residual reduction; the target is 4 orders for the subsonic case and 3 for the supersonic one. Source: `report/limiter_study.json`.

case	limiter treatment	orders	C_D	C_L	status
naca0012_m015_inviscid	Venkatakrishnan, frozen (production default)	4.18	0.0009289	2.95e-04	converged
	Venkatakrishnan, re-evaluated every step	4.00	0.0009075	3.51e-04	converged
	Barth–Jespersen, re-evaluated every step	1.89	0.0005304	3.86e-04	failed
	no limiter	4.00	0.0003509	1.16e-03	converged
naca0012_m200_inviscid	Venkatakrishnan, frozen (production default)	3.87	0.08813	-4.06e-03	converged
	Venkatakrishnan, re-evaluated every step	1.34	0.08818	-4.05e-03	failed
	Barth–Jespersen, re-evaluated every step	-0.54	0.08705	-5.05e-03	failed
	no limiter	-16.81	0.09181	2.66e-03	failed

Three things follow from that table. First, freezing is not what makes the smooth subsonic case work: with the exact wall-pressure Jacobian in place the live Venkatakrishnan limiter converges there too, so freezing is not papering over a defect in the smooth-flow discretisation. Second, on the shock-dominated case the live limiter stalls well short of the target and freezing is what reaches it, while the non-differentiable Barth–Jespersen min stalls on the subsonic case and fails to converge at all on the supersonic one. Third, removing the limiter is not an option: it multiplies the spurious lift of the symmetric subsonic case several-fold and diverges outright at $M_\infty = 2$.

The honest consequence is that the production results for the shocked cases are converged with a frozen limiter field, so the operator being converged in the final phase is the limited scheme with a *fixed* limiter rather than the fully nonlinear limited scheme. The two agree closely — on the supersonic case the frozen and live runs differ in C_D by well under one percent and simply converge to very different residual levels — but they are not identical, and the last significant figures of the force coefficients can move. The freeze step is recorded per run as `limiter_frozen_at_step` so this is auditable.

Discretisation error on the supplied meshes is not quantified. The order of accuracy of the spatial operator *is* measured, on analytically generated meshes carrying a manufactured solution (section 4.3), and it behaves as designed. What that study does not give is the discretisation error of these particular body-fitted meshes: only one mesh per geometry is supplied, so no systematic refinement study of the actual cases could be performed and no error bars are quoted on any reported force coefficient. The zero-lift symmetry of the aerofoil cases and the near-zero inviscid drag at $M_\infty = 0.15$ are used as case-level verification signals instead, and the cylinder results are compared with published values, but those are consistency checks rather than convergence measurements.

The supplied meshes are coarse in the far field. The aerofoil mesh carries only 80 far-field edges over a domain of radius 80 chords and the cylinder mesh only 20 edges over a 400×400 box. The characteristic far-field condition handles this well for the steady cases, but for the $Re = 200$ transient the wake leaves the domain through very few, very large faces, which limits how cleanly the outflow can be treated. A far-field vortex correction was not implemented; at zero angle of attack the induced circulation is negligible, so this only matters if the solver were reused at incidence.

The cylinder wake is under-resolved away from the body. The supplied cylinder mesh puts almost all of its 10 185 cells near the cylinder, so the cells grow rapidly downstream and the shed vortices are numerically diffused within a few diameters. The consequence is visible in the $Re = 200$ vorticity figure and, quantitatively, in the Strouhal number, mean drag and lift amplitude, all of which come out below the accepted incompressible values by 5–30%. This is a spatial-resolution limitation of the supplied mesh combined with the numerical dissipation of the Roe flux at these cell sizes, not a defect of the transient scheme: the inner dual-time solve met its 10^{-3} target on every one of the 30 000 physical steps.

Shock capture is monotone but not sharp. Shocks in the $M_\infty = 0.8$ and $M_\infty = 2.0$ cases are captured over roughly two to three cells. The entropy fix applied to the shear/entropy wave family suppresses the carbuncle instability at the bow shock but adds dissipation wherever the normal Mach number is small, which slightly thickens the boundary layer in the viscous supersonic case. A shock-detecting switch that applies the shear fix only near strong compressions would be a straightforward improvement.

The implicit solver uses an approximate Jacobian. The Jacobian in eq. (27) is exact for the inviscid interface split and for the wall pressure flux, but the viscous coupling is modelled by a spectral radius and the reconstruction/limiter contributions are omitted entirely, so the method is not a true Newton iteration and its asymptotic convergence rate is linear rather than quadratic. This is visible in the steady histories as the gradual growth of the number of inner Gauss–Seidel sweeps as the CFL number ramps up. A matrix-free Krylov method preconditioned by the present block LU-SGS would remove that limitation.

Parallel efficiency is bounded by problem size. With roughly 20 800 and 10 200 cells, these meshes give each rank only a few thousand cells at $np = 8$. The halo exchange is neighbour-scoped and the only collectives are scalar reductions, but there are several reductions per inner iteration (the residual norm and the linear-solve convergence test), so latency is a significant fraction of the iteration cost at this size. No attempt was made to overlap communication with computation or to reduce the number of synchronisation points, both of which would matter on larger problems.

Results are not bitwise rank-independent. Each partition-crossing face is evaluated redundantly on both owning ranks from halo-exchanged states, gradients and limiter values. The two evaluations are algebraically exact negatives of one another, but the Roe dissipation sum is accumulated in a different order on the two sides, so they differ in the last bits. Changing the rank count therefore perturbs the results at round-off level; table 9 shows the resulting spread is many orders of magnitude below the discretisation error, but exact reproducibility across rank counts is not claimed.

Thermodynamics and turbulence. Only a calorically perfect gas with constant viscosity is implemented. The gas model is isolated behind a small class and the transport coefficients enter the viscous kernel at a single point, so a Sutherland law or a Cantera-backed mixture is a localised change, but neither is implemented or tested here. There is no turbulence model; all viscous cases are laminar, as the benchmark specifies.

Output volume. Field files are written as ASCII VTK. This is portable and readable by every post-processing tool used here, but it is bulky; for the transient case the snapshot interval is taken from the case file and the resulting files dominate the submission size. A compressed binary appended format would be a simple improvement.

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